Partially Coasean Temporary Layoffs*

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Abstract

Whether employment decisions reflect the joint interests of firms and workers is fundamental to understanding labor market responses to shocks and policies. I study temporary layoffs under U.S. unemployment insurance, where benefits raise workers' gains from layoffs but impose experience-rated costs on firms. These opposing incentives favor fewer layoffs under unilateral decisions but more under joint decisions. Changes in state schedules reveal that benefits increase temporary layoffs, whereas experience rating reduces them. A model combining these responses implies that 12% of marginal decisions are joint: temporary layoffs are only partially Coasean, yet this minority reverses their response to benefits.

JEL Classification: J63, J64, J65
Keywords: Temporary Layoffs; Unemployment Insurance; Experience Rating; Efficient Separations; Wage Rigidity

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1 Introduction

How employment relationships are maintained or terminated in response to economic shocks is a fundamental question in labor economics and macroeconomics. It matters for unemployment fluctuations, the transmission of monetary and productivity shocks, and the effects of labor market policies. In the Coasean view, firms and workers exploit mutually beneficial arrangements, through wage adjustments or other transfers, to maximize their joint surplus (Coase 1960). Employment decisions can therefore maximize joint surplus even when wages are rigid (Barro 1977). An alternative view is that firms make layoff decisions unilaterally at wages that cannot readily be renegotiated. Because firms do not fully internalize workers' payoffs, wage rigidity can lead to layoffs that do not maximize joint surplus. Workhorse models of the labor market take one view or the other: the canonical search model of unemployment builds on the first (Mortensen and Pissarides 1994), whereas models in which rigid wages drive layoffs build on the second (Hall and Lazear 1984; Blanco et al. 2026). Yet little is known about which view better describes actual employment decisions. More fundamentally, to what extent do employment decisions internalize workers' payoffs?

Distinguishing these views empirically is challenging. Both predict layoffs following adverse shocks, and the arrangements through which firms and workers make employment decisions are rarely observed. I address this question through temporary layoffs and unemployment insurance (UI) in the United States. Temporary layoffs allow firms to suspend employment during periods of low demand while retaining the possibility of recalling workers, and they account for about a third of separations into unemployment (Fujita and Moscarini 2017; Gertler et al. 2026). Two features of the UI system create contrasting incentives. First, workers generally receive UI benefits when laid off by their employers but not when they voluntarily quit. Second, experience rating makes employers bear part of the benefits paid to their laid-off workers. Holding other behavioral responses fixed, a higher UI benefit therefore makes a temporary layoff more costly to a firm deciding alone, but more valuable to a firm-worker pair that jointly weighs the worker's payoff. The two views thus predict opposite responses of temporary layoffs to UI generosity. Variation in experience rating provides complementary information about firms' layoff incentives, making it possible to measure the extent to which decisions are joint.

I start by formalizing this idea in a model of the decision to lay off within a match. The wage is set when the match forms and stays fixed afterward, and the firm decides alone by default. The model incorporates temporary match-specific negative shocks to demand or productivity, temporary layoffs with recall, and the rules of the UI system in the U.S. With probability ϕ_+, the firm and the worker can agree on a temporary layoff that the worker wants and the firm does not, and the worker then compensates the firm with a transfer that is not a wage. Joint decisions therefore do

not need the wage to adjust, unlike in models where separations are unilateral until the wage is renegotiated (Blanco et al. 2026). The parameter ϕ_+ measures how often such layoffs are decided jointly, from zero, when the firm always decides alone, to one. What the data reveal is the share of joint decisions at the margin, which weights ϕ_+ by how many shocks lie at each threshold. The firm's and the pair's thresholds for a layoff have closed forms, and they differ only by the worker's gain from a layoff.

The model makes the logic of the measurement precise. The response of temporary layoffs to higher benefits is positive when joint decisions are sufficiently common at the margin. The larger the experience-rating share, the more prevalent joint decisions must be for this response to be positive. Yet the magnitude of the benefit response alone cannot pin down their share: it also depends on how many matches are close to the layoff threshold. The same elasticity could reflect many joint decisions but few shocks near the threshold, or fewer joint decisions but many shocks near it. Experience rating tells the two cases apart. Unlike a higher benefit, which affects the incentives of the firm and the worker differently, a higher experience-rating share imposes an additional cost on the firm without changing the worker's payoff. It therefore makes a temporary layoff less valuable to the firm and to the pair by the same amount, shifting their layoff thresholds equally. Consequently, the response to experience rating measures the density of shocks at the margin, given the model-implied shift in the thresholds. Combining the two responses then eliminates this density and reveals the share of joint decisions at the margin.

I then estimate the two responses, starting with the response to the benefit. I exploit variation in the statutory benefit schedules across states and over time, which reaches workers with different earnings differently, in the spirit of the simulated instruments of Currie and Gruber (1996a,b). I find that a 10% increase in UI benefits increases the transition from employment to temporary layoff by 5.0% in the CPS and by 5.4% in the SIPP, while it has almost no impact on separations to permanent layoff. However, a positive response could arise even without joint decisions, if a higher benefit made laid-off workers search less: the firm would then be less likely to lose its worker to a new employer during the layoff, and a layoff would cost it less. Yet I find little sign of such a change: more generous benefits barely change the rate of reemployment, and the share of workers on temporary layoff who report searching shows no detectable response to the benefit.

Turning to experience rating, I build the experience-rating share from the tax schedules reported by states. Measured by how much employers' tax rates actually respond to their layoffs, the effective share is 0.12: an employer repays about 12 cents on average of each dollar of benefits that its laid-off workers collect. This is about a fifth of what the schedules imply, partly because many employers sit on the minimum or maximum tax rate, where an additional layoff does not raise their taxes. Using only changes in the schedules, a unit increase in the effective share lowers the transition into temporary layoff by 2.31 log points, and it does not affect permanent separations.

To quantify the extent of joint decision-making, I calibrate the model to the estimated responses to UI benefits and experience rating. Under purely unilateral decisions, higher benefits would reduce temporary layoffs, with an elasticity of -0.29, which is the opposite of what the data show. To reproduce the observed positive elasticity through reduced search alone, the share of laid-off workers who search would have to fall by at least 0.45% for every 1% increase in benefits. Yet the estimates show no statistically significant decline in this share. Combining the two estimated responses, the model implies that about 12% of temporary layoff decisions at the margin weigh the worker's payoff. Thus, although most decisions at the margin are unilateral, the unilateral benchmark predicts the wrong sign for their response to UI benefits.

The importance of this minority of joint decisions depends on the shock. For changes in the real wage, the unilateral benchmark provides a much closer approximation. When firms decide alone, a lower real wage reduces their incentive to lay off workers. In contrast, when decisions are joint, the wage largely divides the value of the match without changing the layoff decision, as argued by Barro (1977). To quantify this distinction, I consider a 1% rise in the price level that lowers the real wage of existing matches, with wages renegotiated at the frequency measured by Grigsby et al. (2021). In the calibrated economy, the transition into temporary layoff falls by 21% on impact, compared with 24% under purely unilateral decisions. In log points, the response is about 89% of the unilateral benchmark, close to the fraction of marginal decisions that are not joint, and this ratio is largely insensitive to the persistence of the shock. Thus, whether unilateral decision-making provides a useful approximation depends not only on how prevalent joint decisions are, but also on how a shock affects the incentives of firms and workers.

Related literature. This paper is most closely related to two recent papers on whether separations are Coasean. Jäger et al. (2023) test the Coasean view with an extension of UI benefits in Austria: after its repeal, the jobs that had survived it were no more robust than other jobs, which rejects efficient separations among older workers. Blanco et al. (2026) build an equilibrium model in which wages are fixed between staggered renegotiations and quits and layoffs are unilateral, and they characterize how such separations shape the response of the labor market to inflationary shocks. Relative to these papers, I make two contributions by focusing on temporary layoffs and exploiting the eligibility rules and experience rating of UI in the U.S. First, I measure how Coasean decisions are rather than test whether they are, and I find that they are mostly unilateral, but that the minority of joint decisions reverses their response to UI benefits. Second, I allow decisions to be joint even though the wage is rigid. The firm and the worker can settle a layoff through a transfer that is not a wage, in the spirit of Barro (1977), whereas in Blanco et al. (2026) separations are unilateral as long as the wage is not renegotiated. The degree is then a property of the allocation, which I measure from layoff decisions instead of wage changes. For the inflationary shocks that Blanco

et al. (2026) study, the unilateral benchmark is a close approximation: in the calibrated model, the response of temporary layoffs is about 89% of it. The findings also complement those of Jäger et al. (2023): they study separations that end the relationship, and I study separations after which the relationship continues.

The broader question goes back to Barro (1977), who argued that long-term contracts separate the efficiency of employment from the flexibility of wages, and to Hall (1980), who described employment relationships in which the pair adjusts to temporary declines in demand efficiently and takes UI into account. Hall and Lazear (1984) showed that fixed prices lead to inefficient quits and layoffs when the parties cannot renegotiate, and McLaughlin (1991) built a theory of quits and layoffs with efficient turnover. Earlier tests of efficient bargaining in unionized markets asked whether employment lies on the contract curve or on the labor demand curve (MaCurdy and Pencavel 1986; Brown and Ashenfelter 1986; Card 1986). Davis and Krolikowski (2025) find that wages are rigid on the layoff margin.

The theory of temporary layoffs under UI dates back to Feldstein (1976) and Baily (1977), followed by, among others, Burdett and Wright (1989a,b). These papers assume that the firm and the worker decide together. I nest their decision rule with the unilateral one and let the data tell how much weight each receives. The empirical literature on the effect of UI on job separations is thin compared to the literature on job finding (Le Barbanchon et al. 2024). Feldstein (1978) uses cross-sectional variation in UI benefits and finds a positive correlation between temporary layoff unemployment and UI generosity, Topel (1984) finds that UI encourages spells of unemployment because experience rating is incomplete, and Saffer (1982, 1983) relates layoffs, and then temporary and permanent layoffs separately, to UI. More recent papers exploit plausibly exogenous variation to estimate the effect of UI on separations to unemployment (Tuit and van Ours 2010; Lalive et al. 2011; Albanese et al. 2020; Jäger et al. 2023; Hartung et al. 2026), without a distinction between temporary and permanent layoffs. I provide causal evidence for temporary and permanent layoffs separately, and evidence on what laid-off workers do during the layoff.

The model is related to search and matching models with temporary layoffs and recalls (Fujita and Moscarini 2017; Lam and Qiu 2024; Albertini et al. 2023; Gertler et al. 2026), but I hold the labor market fixed and focus on who decides a temporary layoff. In a broad sense, this paper also contributes to the literature on social insurance and labor market policies that emphasizes how firms respond to them (Acemoglu and Shimer 1999; Pries and Rogerson 2005; Blanchard and Tirole 2008; Jung and Kuester 2015; Mitman and Rabinovich 2015; Landais et al. 2018a,b; Aizawa et al. 2025, 2026).

2 Model

This section builds a model of how temporary layoffs are decided, and Section 3 derives what the response of temporary layoffs to unemployment insurance reveals about it.

The model describes a single match. It takes a wage that is set when the match forms and stays fixed afterward, and a firm that decides alone when to stop production. I add temporary negative shocks to demand, temporary layoffs with recall, the rules of U.S. unemployment insurance, and a parameter that measures how often the decision to lay off is made jointly by the firm and the worker. I treat permanent separations as exogenous, focus on the choice between production and temporary layoff, and hold the labor market outside the match fixed: the wage, the rate at which jobless workers find jobs, and the rate at which workers on layoff find other jobs are given. Section 2.2 discusses these assumptions.

2.1 Environment

Time is continuous. There is a unit measure of workers, and workers and firms are risk neutral and discount the future at rate r.

Matches and output. A firm and a worker form a match, which produces a flow of output y in normal times. At rate λ_d, a match in normal times is hit by a temporary match-specific negative shock to demand or productivity. During a shock, the match can produce $y - \varepsilon$, where the size $\varepsilon \geq 0$ is drawn from a distribution G with density g when the shock arrives and stays fixed until the shock ends, which happens at rate λ_r.

Temporary layoffs and permanent separations. During a shock, the match either produces or stops production with the worker on temporary layoff. A worker on temporary layoff remains attached to the firm and is recalled when the shock ends. A match ends for two reasons. It is destroyed at rate δ whatever its state, after which the worker is jobless. I call this a permanent separation and treat it as exogenous. A worker on temporary layoff may also find a job elsewhere, which ends the match. I call this a move to a new employer. For a worker on temporary layoff, either event ends the prospect of a recall, and I call both a loss of recall, extending the term that Gertler et al. (2026) use for workers on temporary layoff who lose their jobs permanently.

Unemployment insurance. A worker who is not working, on temporary layoff or jobless, enjoys a flow value of non-work time ℓ and collects a flow benefit b. A worker who quits does not collect the benefit. Experience rating charges a firm a share $e \in [0, 1]$ of the benefits its own laid-off workers collect. The firm pays the flow eb while its worker is on temporary layoff, and

$c = eb/(r + f)$ when the match is destroyed, a share e of the discounted benefits that the jobless worker collects until finding a job at the rate f defined below. The benefits not charged to firms are financed by a lump-sum tax on all workers. In the model, every worker who is not working collects the benefit; Section 5.1 discusses how the data, in which many workers collect less, map into the model.

Wages and the labor market outside the match. The wage w is set when the match forms and stays fixed for the life of the match, including after a recall.[1] The wage of an existing match therefore does not respond to a change in the benefit.[2] A jobless worker finds a job at rate f, at the same wage. A worker on temporary layoff moves to a new employer at rate s, which I call the rate of moving to a new employer. If laid-off workers search less when the benefit is higher, s falls with the benefit, and I write its elasticity with respect to the benefit, signed so that it is positive when s falls, as $\eta_{s,b} = -\partial \log s / \partial \log b$. I take w, f, and s as given, and write $\tilde{W}_0$ for the value of the job that a worker on temporary layoff moves to, which Section 2.4 specifies.

Who decides a temporary layoff. Under U.S. rules a temporary layoff is the employer's decision: a worker cannot stop working and collect the benefit, because a worker who quits is generally not entitled to it. I therefore let the firm choose between production and temporary layoff to maximize its own value, taking the wage as given. This is the unilateral decision of Blanco et al. (2026), applied to the margin between production and temporary layoff.

Agreements. The worker's payoff can still enter the decision through an agreement with the firm. When a shock arrives, with probability ϕ_+ the pair can agree on a temporary layoff that the worker prefers and the firm does not. If the worker prefers a temporary layoff to working and the joint value of the layoff exceeds that of production, the worker is laid off and makes up the firm's loss, for example through flexibility in hours or tasks after the recall. With probability ϕ_- the pair can instead agree to keep producing when the firm prefers a layoff and the worker prefers to work; I call this an agreement to keep working. In both cases the agreement covers the whole shock, and the compensation is the firm's loss at the start of the shock. I treat the compensation as a transfer that does not depend on whether the worker is later recalled, so that the firm is exactly compensated, the worker keeps the gain, and an agreement maximizes the joint value of the pair. In the spirit of Barro (1977), the wage stays fixed, and the transfer lets the pair carry out the decision that maximizes its joint value. When no agreement is available, the firm's choice stands.

[1] In the SIPP, Lam and Qiu (2024) find that the wage changes of workers recalled after a jobless spell are concentrated at zero, and that wages remain largely unchanged before and after a recall for workers of all ages.

[2] This is consistent with Jäger et al. (2020), who find that the wages of employed workers in Austria rise by less than one cent per dollar of benefit.

The parameters ϕ_+ and ϕ_- measure how often the decision is joint, that is, Coasean; their subscripts refer to the sign of the worker's gain from a layoff, defined in Section 2.4. The unilateral benchmark is $\phi_+ = \phi_- = 0$, and the joint decision is $\phi_+ = \phi_- = 1$. Which of the two can move a decision depends on whether the worker prefers a temporary layoff to working (Section 2.4), and the responses in the data measure the share of joint decisions at the margin, which combines the two (Section 3).

2.2 Discussion of the Main Assumptions

Permanent separations. Permanent separations do not respond to the benefit in the data (Section 4.3), and I therefore treat them as exogenous and focus on the choice between production and temporary layoff.

The labor market outside the match. Holding the wage and the job-finding rate fixed is the comparison the empirical design makes, as long as the earnings bins and sectors of a state share its labor market; the rate of moving to a new employer is also fixed unless laid-off workers search less when their own benefit is higher, which Section 4.5 examines. The response to experience rating compares sectors within a state and year, and the benefit elasticity in the CPS is about the same when it compares earnings bins within a state and month (column (3) of Table 1), so that the responses are those of matches that face the same labor market. Section 3.1.4 discusses how the response of the market and of the wages of new matches could add to the responses of the model, and Section 4.5 discusses the evidence on both.

The two kinds of agreements. Agreements to lay off and agreements to keep working need not be equally frequent. In an agreement to keep working, the worker gives up pay to cover the firm's loss during the shock, either at once through a pay cut or later in the relationship, and a pay cut meets the same frictions that make wages rigid. In the survey of UI recipients by Davis and Krolikowski (2025), between 34% and 55% of the workers on temporary layoff, depending on the size of the proposed cut, would have accepted a temporary pay cut to avoid the layoff, yet only 4.2% report that a cut in pay, benefits, or hours was discussed as an alternative to the layoff. An agreement to lay off asks nothing of the worker's pay while working, and the compensation can be delivered later in the relationship, as in the long-term arrangements described by Hall (1980).

The transfer. An agreement pays the firm exactly its loss and leaves the whole gain to the worker. The two thresholds then do not depend on the probabilities of agreements, which only decide which threshold applies. If the firm kept part of the gain, agreements would raise the firm's value of a match on layoff, and with it the firm's threshold. I also treat the transfer as owed whatever happens

later, which abstracts from the risk that the match ends before the worker makes up the loss. The share of joint decisions at the margin is defined under these assumptions.

2.3 Values

I first write the values of the firm and the worker for a given decision rule, which Section 2.4 then derives. Since the size of a shock is fixed while it lasts, the decision is the same throughout a shock.

Firms. Let J_0 denote the firm's value of a match in normal times. The firm's values of producing through a shock of size ε and of a temporary layoff through it satisfy the following Hamilton-Jacobi-Bellman (HJB) equations

$$rJ_E(\varepsilon) = y - \varepsilon - w + \lambda_r\big(J_0 - J_E(\varepsilon)\big) - \delta\big(J_E(\varepsilon) + c\big), \tag{1}$$

$$rJ_T = -eb + \lambda_r\big(J_0 - J_T\big) - sJ_T - \delta\big(J_T + c\big). \tag{2}$$

On temporary layoff, the firm pays the experience-rated charge, regains the match when the shock ends, and loses it if the worker finds another job. The firm's value at the start of a shock is $J(\varepsilon) = \max\{J_E(\varepsilon), J_T\}$ whether or not an agreement moves the decision, because the firm is compensated in an agreement. In normal times, the value satisfies

$$rJ_0 = y - w + \lambda_d \int \big(J(\varepsilon') - J_0\big) dG(\varepsilon') - \delta\big(J_0 + c\big). \tag{3}$$

Workers. Let W_0 denote the value of a worker in a match in normal times and U the value of a jobless worker. A worker who produces through a shock, or who is on temporary layoff through it, has the values

$$rW_E = w + \lambda_r\big(W_0 - W_E\big) - \delta\big(W_E - U\big), \tag{4}$$

$$rW_T = b + \ell + \lambda_r\big(W_0 - W_T\big) + s\big(\tilde{W}_0 - W_T\big) - \delta\big(W_T - U\big). \tag{5}$$

In normal times,

$$rW_0 = w + \lambda_d\Big((1 - P)\, W_E + P\, W_T - M - W_0\Big) - \delta\big(W_0 - U\big), \tag{6}$$

where P is the probability that a shock leads to a temporary layoff, defined in Section 2.4, and M is the payment that the worker expects to make when a shock arrives, since in an agreement the

worker pays the firm its loss from the decision, $|J_E(\varepsilon) - J_T|$.[3] A jobless worker collects the benefit, enjoys the value of non-work time, and finds a job at rate f,

$$rU = \ell + b + f\big(W_0 - U\big). \tag{7}$$

2.4 The Decision Rule

The flow gains from a temporary layoff. Subtracting equation (1) from equation (2) gives the firm's gain,

$$(r + \delta + \lambda_r)\big(J_T - J_E(\varepsilon)\big) = \Delta_F(\varepsilon), \tag{8}$$

where

$$\Delta_F(\varepsilon) \equiv w - (y - \varepsilon) - eb - sJ_T. \tag{9}$$

$\Delta_F(\varepsilon)$ represents the firm's flow gain from putting the worker on temporary layoff rather than producing through a shock of size ε. Equation (8) states that the firm prefers a temporary layoff to production exactly when $\Delta_F(\varepsilon) > 0$. A temporary layoff saves the wage net of output and of the experience-rated charge, and it puts the match at risk: the firm's value of a match on layoff, J_T, is lost at rate s. The comparison is between flows, and J_T does not depend on the size of the shock. Subtracting equation (4) from equation (5) gives the worker's gain,

$$(r + \delta + \lambda_r)\big(W_T - W_E\big) = \Delta_W, \tag{10}$$

where

$$\Delta_W = b + \ell - w + s\big(\tilde{W}_0 - W_T\big). \tag{11}$$

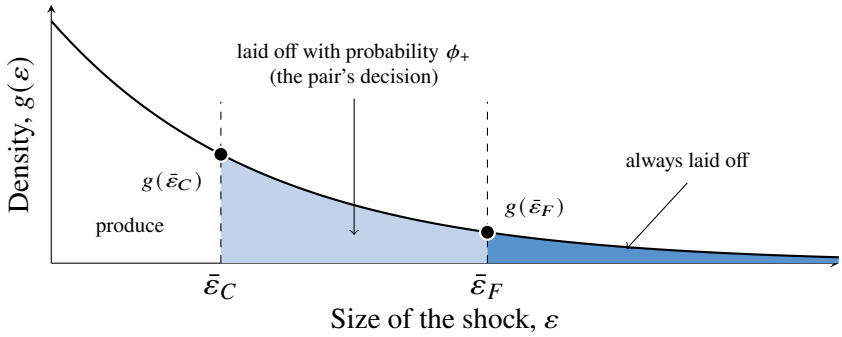
To keep the thresholds in closed form, I assume that the new job is worth as much as waiting on layoff for the recall, $\tilde{W}_0 = W_T$, so that a move to a new employer costs the pair only the firm's value of the match and $\Delta_W = b + \ell - w$.[4] Δ_W represents the worker's flow gain from a temporary layoff, the benefit and the value of non-work time net of the wage; it does not depend on the size of the shock, because the worker's continuation values are the same under both choices. The joint gain is

[3] For example, in an agreement to lay off, the worker can make up the firm's loss after the recall through flexibility in hours or tasks. In an agreement to keep working, the worker can do the same later in the relationship, or take a temporary pay cut.

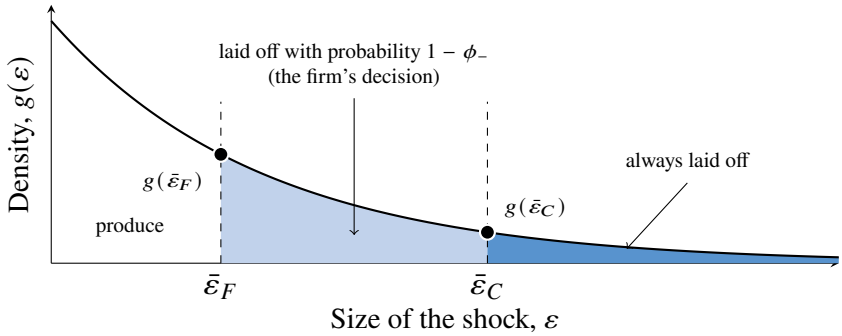
[4] A worker moves only to a job that is at least as good as waiting, and two facts suggest that such jobs are not much better. Workers who move to a new employer after a jobless spell earn less than before on average, by about 3% at ages 23 to 35 and by more than 10% at ages 35 to 60, whereas recalled workers keep their wage (Lam and Qiu 2024). And 57% of the workers on temporary layoff in the CPS did not look for work in the last four weeks (Table 4). Section 6 returns to this assumption when the wage of a match differs from the wage at which new matches form.

Figure 1: The Decision to Lay Off

(a) Positive-Gain Match, $b + \ell > w$: An Agreement to Lay Off Can Move the Decision



(b) Negative-Gain Match, $b + \ell < w$: An Agreement to Keep Working Can Move the Decision



Notes: Stylized density of shock sizes. The firm lays off when $\varepsilon > \bar{\varepsilon}_F$, and the pair prefers a layoff when $\varepsilon > \bar{\varepsilon}_C$. Between the two thresholds the decision depends on whether an agreement is available. The dots mark the density at the two thresholds, which sets the share of joint decisions at the margin, $\tilde{\phi}$, in Section 3.

the sum of the two,

$$\begin{aligned} \Delta_C(\varepsilon) &= \Delta_F(\varepsilon) + \Delta_W \\ &= (1 - e)b + \ell - (y - \varepsilon) - sJ_T. \end{aligned} \tag{12}$$

$\Delta_C(\varepsilon)$ represents the joint flow gain of the firm and the worker from a temporary layoff, the gain that a Coasean decision weighs. Equation (12) shows that the wage drops out of the flows in the joint gain, because it is a transfer within the pair, and enters it only through the expected loss of the match, sJ_T. It also shows that the benefit enters the joint gain net of the experience-rated charge, as a subsidy of $1 - e$ per unit. In the firm's gain (8), by contrast, the benefit enters only through the charge.

The decision rule. Both gains increase one for one in ε. The firm therefore lays off when $\varepsilon > \bar{\varepsilon}_F$ and the pair prefers a layoff when $\varepsilon > \bar{\varepsilon}_C$, where

$$\bar{\varepsilon}_F = (y - w) + eb + sJ_T, \qquad \bar{\varepsilon}_C = \bar{\varepsilon}_F - (b + \ell - w). \tag{13}$$

Equation (13) states that the firm lays off when the shock exceeds the sum of the profit margin, the experience-rated charge, and the expected loss of the match, and that the pair's threshold differs from the firm's only by the worker's flow gain Δ_W. I call a match a positive-gain match when its worker gains from a temporary layoff, $\Delta_W = b + \ell - w > 0$, and a negative-gain match otherwise.[5] Figure 1 shows the decision in the two types. In a positive-gain match, panel (a), $\bar{\varepsilon}_C < \bar{\varepsilon}_F$ and an agreement to lay off can move the decision; in a negative-gain match, panel (b), $\bar{\varepsilon}_C > \bar{\varepsilon}_F$ and only an agreement to keep working can. Let ϕ denote the probability of the agreement that can move the decision, ϕ_+ in a positive-gain match and ϕ_- in a negative-gain match. In either type, a shock leads to a temporary layoff when it exceeds the firm's threshold and no agreement is available, or when it exceeds the pair's threshold and an agreement is available. Shocks above both thresholds therefore always lead to a layoff, and shocks between them lead to one with probability ϕ_+ in a positive-gain match and $1 - \phi_-$ in a negative-gain match. The probability that a shock leads to a temporary layoff is

$$P = \underbrace{(1 - \phi)\left(1 - G(\bar{\varepsilon}_F)\right)}_{\text{Unilateral}} + \underbrace{\phi\left(1 - G(\bar{\varepsilon}_C)\right)}_{\text{Coasean}}, \tag{14}$$

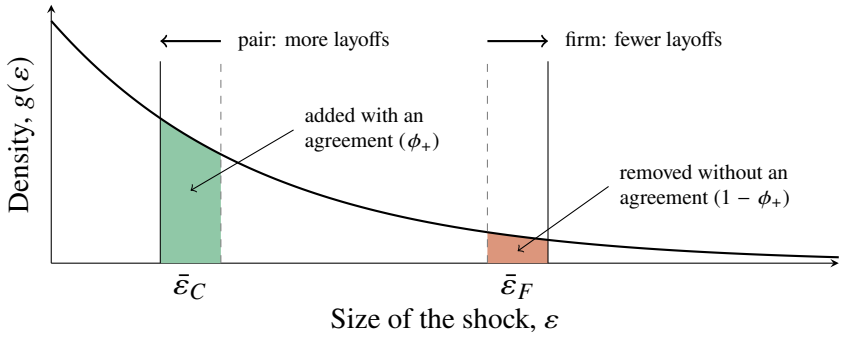
and matches in normal times flow into temporary layoff at rate $\lambda_d P$. Equation (14) states that the probability of a temporary layoff mixes the unilateral and the Coasean decision rules, with weights $1 - \phi$ and ϕ. Given the wage, the job-finding rate, and the rate of moving to a new employer, the firm's values (1)–(3) and the thresholds (13) determine the decisions, and the worker's values (4)–(7) then follow.

3 What the Response to the Benefit Reveals

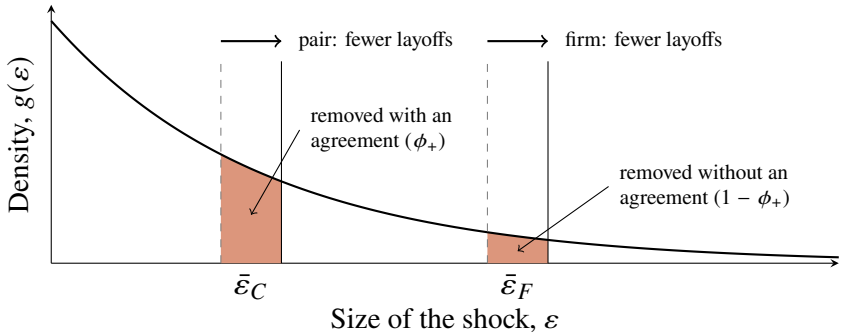
This section derives how the benefit and other policies move temporary layoffs in the model of Section 2, and what the response to the benefit reveals about who decides.

Figure 2: How Policies Move the Thresholds in a Positive-Gain Match

(a) A Higher Benefit: The Two Thresholds Move Apart



(b) Experience Rating or a Subsidy to Production: The Two Thresholds Move Together



Notes: Stylized density of shock sizes. Dashed lines are the thresholds before the change, and solid lines after it. Panel (a) is drawn for $e > R_b$, so that the firm alone lays off less when the benefit rises. The response of temporary layoffs is the mass added in green net of the mass removed in red, each weighted by the probability that applies.

3.1 How the Benefit Moves Temporary Layoffs

Consider a match of type $k \in \{+, -\}$, positive-gain or negative-gain, and write P_k and ϕ_k for its P and ϕ in equation (14). The probability P_k moves with a policy only through the two thresholds. A policy x shifts the firm's and the pair's thresholds by $\partial\bar{\varepsilon}_F/\partial x$ and $\partial\bar{\varepsilon}_C/\partial x$, and a lower threshold means more layoffs. Differentiating equation (14), the log change in the probability of a temporary layoff in response to a policy change ∂x is

$$\frac{\partial \log P_k}{\partial x} = -\bar{h}_k \left((1-\tilde{\phi}_k)\, \frac{\partial \bar{\varepsilon}_F}{\partial x} + \tilde{\phi}_k\, \frac{\partial \bar{\varepsilon}_C}{\partial x} \right), \tag{15}$$

where

$$\bar{h}_k = \frac{(1-\phi_k)\, g(\bar{\varepsilon}_F) + \phi_k\, g(\bar{\varepsilon}_C)}{P_k}, \qquad \tilde{\phi}_k = \frac{\phi_k\, g(\bar{\varepsilon}_C)}{(1-\phi_k)\, g(\bar{\varepsilon}_F) + \phi_k\, g(\bar{\varepsilon}_C)}, \tag{16}$$

[5] Both types appear in the survey of UI recipients by Davis and Krolikowski (2025): the workers on temporary layoff who would have accepted a temporary pay cut to avoid the layoff preferred working to the layoff, as in a negative-gain match (Section 2.2).

and I take the density g to be positive at both thresholds. Equation (15) states that every policy moves temporary layoffs through the same two objects: the density of shocks at the margin, $\bar{h}_k$, which scales the response, and the share of joint decisions at the margin, $\tilde{\phi}_k$, which weights the shifts of the two thresholds. Figure 2 illustrates this in a positive-gain match: a policy that moves a threshold adds or removes the shocks next to it, and those at the pair's threshold count with the probability of an agreement, ϕ_+, and those at the firm's threshold with $1 - \phi_+$. The factor $\bar{h}_k$ is the density of shocks at the two thresholds (the dots in Figure 1) relative to the probability of a layoff, and it measures how many shocks lie at the margin. The weight $\tilde{\phi}_k$ is the share of joint decisions in the density of shocks at the margin. It is not the share of observed layoffs that are joint, and since the two terms in equation (15) can have opposite signs, it is not the share of the response either. The weight exceeds ϕ_k if and only if $g(\bar{\varepsilon}_C) > g(\bar{\varepsilon}_F)$. When the density of shock sizes falls with their size, this holds in a positive-gain match, where the pair's threshold lies below the firm's, where shocks are more common, and the more so the larger the worker's gain from a layoff; in a negative-gain match, where the pair's threshold lies above the firm's, $\tilde{\phi}_- < \phi_-$. Both $\bar{h}_k$ and $\tilde{\phi}_k$ depend on where the thresholds are, and the results of this section are local. For a small change from the current policy, however, the same $\bar{h}_k$ and $\tilde{\phi}_k$ apply whichever policy changes, which is what makes the response to the benefit informative about the response to other policies.

The shifts in the thresholds. By equation (13), a policy moves the thresholds through the flows in the gains and through the expected loss of the match, sJ_T, which is the same in both. For the benefit, in matches with a given wage,

$$\frac{\partial \bar{\varepsilon}_F}{\partial b} = e - R_b, \qquad \frac{\partial \bar{\varepsilon}_C}{\partial b} = -(1 - e) - R_b, \qquad R_b \equiv -\frac{\partial (sJ_T)}{\partial b}. \tag{17}$$

Equation (17) implies that the benefit moves the two thresholds in opposite directions through the static terms and in the same direction through the common term R_b, as panel (a) of Figure 2 illustrates for $e > R_b$. The static terms say that a higher benefit raises the firm's flow cost of a temporary layoff by e, and the pair's flow gain by the subsidy $1 - e$. The common term R_b is the fall in the expected loss of the match. It has two parts, $R_b = \eta_{s,b}\, sJ_T/b - s\, dJ_T/db$, where $\eta_{s,b} = -\partial \log s / \partial \log b$ and dJ_T/db is the total change in the firm's value of a match on layoff. The first is retention: a higher benefit lowers the rate of moving to a new employer if the worker searches less. The second is the firm's surplus in the match, J_T. Holding s fixed, the value of a match on layoff changes by

$$(r + \delta + \lambda_r + s)\, \frac{\partial J_T}{\partial b} = -e - \delta\, \frac{\partial c}{\partial b} + \lambda_r\, \frac{\partial J_0}{\partial b}. \tag{18}$$

Equation (18) implies that a higher benefit lowers the firm's value of a match on layoff through the charges during the layoff, the charges on future permanent separations, and the value of the match after the recall.

3.1.1 Unilateral Decisions

Suppose the firm decides alone, $\phi_+ = \phi_- = 0$. At the firm's threshold, equation (13) is an identity between the expected loss of the match and the flow saved by the layoff,

$$sJ_T = w - (y - \bar{\varepsilon}_F) - eb. \tag{19}$$

Equation (19) states that, at the firm's threshold, the expected loss of the match equals the flow that the layoff saves. Retention therefore matters in proportion to this flow, rather than to the size of the firm's surplus in the match.

Proposition 1 (Unilateral Decisions)**.** *Suppose $\phi_+ = \phi_- = 0$ and $e > 0$, and let $\vartheta = -s\,(dJ_T/db)/e$ denote the share of the experience-rated charge offset by the fall in the firm's surplus. A higher benefit raises temporary layoffs if and only if $R_b > e$. When $s > 0$ and $J_T > 0$, this condition is equivalent to*

$$\eta_{s,b} > (1 - \vartheta)\,\frac{e\,b}{sJ_T}, \tag{20}$$

where, by equation (19), the denominator is the flow that a layoff saves at the firm's threshold. Without moves to new employers ($s = 0$), a higher benefit lowers temporary layoffs for any distribution of shocks.

Proof. By equations (15) and (17) with $\phi_k = 0$, $\partial \log P_k/\partial b = \bar{h}_k\,(R_b - e)$. By the definitions of $\eta_{s,b}$ and ϑ, $R_b = \eta_{s,b}\,sJ_T/b + \vartheta e$, so that $R_b > e$ if and only if $\eta_{s,b}\,sJ_T/b > (1 - \vartheta)\,e$; dividing by $sJ_T/b > 0$ gives condition (20). With $s = 0$, $R_b = 0$ and $\partial \log P_k/\partial b = -\bar{h}_k\,e < 0$. □

Proposition 1 says that a firm deciding alone treats a higher benefit as a cost, the charge of e per unit of benefit while its worker is on layoff. The only force in the other direction is that a higher benefit lowers the expected loss of the match during the layoff, either because the worker searches less and is less likely to move to a new employer, or because the firm's surplus in the match falls. Unless the fall in the firm's surplus offsets the whole charge, $\vartheta \geq 1$, a positive response therefore requires the rate at which laid-off workers move to new employers to fall with the benefit by enough to offset the charge, condition (20), which Section 4.5 examines through their search.

3.1.2 Joint Decisions

In the pair's gain (12), the benefit enters net of the charge, as a subsidy of $1 - e$ per unit, and the expected loss of the match is the same as in the firm's gain. The two shifts in equation (17) therefore differ by exactly one, in both types of match. In a positive-gain match, a higher benefit makes an agreement to lay off more valuable to the worker; in a negative-gain match, it makes an agreement to keep working less valuable to the worker, because working pays less relative to the layoff. In either type, the worker's payoff pushes temporary layoffs up when the benefit rises.

Proposition 2 (The Coasean Cutoff)**.** *In a match of either type with a given wage, temporary layoffs rise with the benefit if and only if the share of joint decisions at the margin exceeds a cutoff,*

$$\tilde{\phi}_k > \tilde{\phi}^* \equiv e - R_b. \tag{21}$$

Without moves to new employers, the cutoff is the experience-rating share, $\tilde{\phi}^* = e$.

Proof. By equation (17), $-(1 - \tilde{\phi}_k)\, \partial \bar{\varepsilon}_F / \partial b - \tilde{\phi}_k\, \partial \bar{\varepsilon}_C / \partial b = \tilde{\phi}_k - (e - R_b)$, and the factor $\bar{h}_k$ in equation (15) is positive. □

Proposition 2 makes precise what the sign of the response reveals: a positive response means that joint decisions carry more than the share $\tilde{\phi}^*$ of the margin. Without moves to new employers, a unilateral decision loses e per unit of benefit and a joint decision gains $1 - e$, so that a positive response requires the joint decisions to carry more than the share e of the margin. Retention and the fall in the firm's surplus lower the expected loss of the match in both decisions alike, and they lower the cutoff one for one; Section 5 reports its value at the measured experience rating.

3.1.3 What the Benefit Elasticity Reveals about Other Policies

Equation (15) holds for any policy with the same weight $\tilde{\phi}_k$. A policy that acts on the firm, such as experience rating or a subsidy for producing during a shock, does not enter the worker's flow gain $b + \ell - w$. By equation (13) it moves the two thresholds by the same amount, $\partial \bar{\varepsilon}_F / \partial x = \partial \bar{\varepsilon}_C / \partial x$.

Proposition 3 (Firm-Side Policies)**.** *Let $\eta_{P_k,b} = \partial \log P_k / \partial \log b$ in a match of type k with a given wage, and suppose $\tilde{\phi}_k \neq \tilde{\phi}^*$. For any policy x that acts on the firm,*

$$\frac{\partial \log P_k}{\partial x} = -\frac{\eta_{P_k,b}}{b} \cdot \frac{\partial \bar{\varepsilon}_F / \partial x}{\tilde{\phi}_k - \tilde{\phi}^*}. \tag{22}$$

Without moves to new employers, experience rating raises both thresholds by $\partial \bar{\varepsilon}_F / \partial e = b$, and a flow subsidy τ paid to the firm while it produces during a shock raises both by $\partial \bar{\varepsilon}_F / \partial \tau = 1$, which

gives

$$\frac{\partial \log P_k}{\partial e} = -\frac{\eta_{P_k,b}}{\tilde{\phi}_k - e}, \qquad \frac{\partial \log P_k}{\partial \tau} = -\frac{\eta_{P_k,b}}{(\tilde{\phi}_k - e)\, b}. \tag{23}$$

Proof. Divide equation (15) for x by its counterpart for b, use $-(1-\tilde{\phi}_k)\,\partial\bar{\varepsilon}_F/\partial b - \tilde{\phi}_k\,\partial\bar{\varepsilon}_C/\partial b = \tilde{\phi}_k - \tilde{\phi}^*$, and multiply by $\eta_{P_k,b}/b$. □

Proposition 3 turns the measured benefit elasticity into the effect of any policy that acts on the firm, except at the cutoff, $\tilde{\phi}_k = \tilde{\phi}^*$, where the elasticity is zero whatever the density at the margin. The benefit moves the two thresholds in opposite directions, whereas policies that act on the firm move them together (panel (b) of Figure 2). The measured elasticity therefore translates into the effect of a firm-side policy at a rate set by the distance between the share of joint decisions and the cutoff. Given the measured elasticity, the effect of experience rating is smallest when every decision is joint, $\tilde{\phi}_k = 1$, since neither $\partial\bar{\varepsilon}_F/\partial e$ nor $\tilde{\phi}^*$ depends on ϕ_+ or ϕ_- in matches with a given wage. Without moves to new employers, moving to perfect experience rating, a change of $1 - e$, then lowers temporary layoffs by about $\eta_{P_k,b}$ log points, which is the elasticity restated in other units.[6] With fewer joint decisions, the same elasticity implies a larger effect of experience rating, because it must come from a margin that is denser. In this sense, without moves to new employers, the elasticity alone gives a lower bound on the power of firm-side policies, and $\tilde{\phi}_k$ turns the bound into a number.

3.1.4 The Market and the Wages of New Matches

Two responses outside the match are held fixed here. A higher benefit can lower vacancies, and with them the job-finding rate and the rate of moving to a new employer, which would make a temporary layoff less risky for the firm. It can also move the wage at which new matches form, and a higher wage would lower the firm's threshold and raise temporary layoffs even under unilateral decisions. Section 4.5 discusses the evidence on both.

3.2 Toward Measurement

The model gives three predictions for the data, and one way to measure how Coasean the decisions are.

First, if the firm decides alone, the benefit enters the layoff decision only through the experience-rated charge and the expected loss of the match. A higher benefit then raises temporary layoffs only

[6] The statement applies the local result to a finite change and holds to first order for a general distribution of shock sizes. With exponential shock sizes, as in Section 5, the log probability of a layoff is linear in e when every decision is joint and $s = 0$, and the statement is exact.

if the fall in the expected loss of the match, through retention and the firm's surplus, outweighs the charge (Proposition 1). Without moves to new employers it always lowers them.

Second, if the firm and the worker decide together, the benefit net of the charge is a subsidy to the pair, and a higher benefit raises temporary layoffs, as in Feldstein (1976). Between the two, the response is positive if and only if the share of joint decisions at the margin exceeds the cutoff $\tilde{\phi}^* = e - R_b$ (Proposition 2).

Third, permanent separations do not respond to the benefit in the model by construction, and the behavior of workers on layoff matters only through the rate of moving to a new employer. Section 4 tests these features directly.

Matches of both types. The data pool matches of both types, and the measured responses add up the responses of all matches. Let P now denote the probability that a shock leads to a temporary layoff across all matches, and ω_k the share of temporary layoffs that come from matches of type k. If a policy shifts the thresholds by the same amounts in all matches, as the benefit and experience rating do without moves to new employers when the matches share the benefit, equation (15) holds for P,

$$\frac{\partial \log P}{\partial x} = -\bar{h}\left((1-\tilde{\phi})\,\frac{\partial \bar{\varepsilon}_F}{\partial x} + \tilde{\phi}\,\frac{\partial \bar{\varepsilon}_C}{\partial x}\right), \qquad \bar{h} = \sum_k \omega_k\,\bar{h}_k, \qquad \tilde{\phi} = \sum_k \frac{\omega_k\,\bar{h}_k}{\bar{h}}\,\tilde{\phi}_k. \tag{24}$$

Equation (24) states that, across all matches, the density at the margin is the average of $\bar{h}_k$ weighted by the shares of temporary layoffs, and the share of joint decisions at the margin is the average of $\tilde{\phi}_k$ weighted by each type's contribution to that density. This share combines agreements to lay off and agreements to keep working.

Measuring the share of joint decisions. The response to the benefit alone does not determine the share of joint decisions, because it also depends on how many shocks lie at the margin. Let $\eta_{P,b} = \partial \log P / \partial \log b$. By equation (24),

$$\eta_{P,b} = b\,\bar{h}\,(\tilde{\phi} - \tilde{\phi}^*), \qquad \frac{\partial \log P}{\partial e} = -\bar{h}\,\frac{\partial \bar{\varepsilon}_F}{\partial e}. \tag{25}$$

Equation (25) implies that the benefit elasticity mixes the share of joint decisions with the density of shocks at the margin, $\bar{h}$, whereas the response to experience rating, a policy that moves both thresholds alike, measures the density alone. Together, the two responses give

$$\tilde{\phi} = \tilde{\phi}^* + \frac{\eta_{P,b}\,\partial \bar{\varepsilon}_F / \partial e}{b\,|\partial \log P / \partial e|}. \tag{26}$$

Equation (26) states that the share of joint decisions at the margin equals the cutoff plus the ratio of the two measured responses, scaled by $(\partial\bar{\varepsilon}_F/\partial e)/b$. Apart from $\tilde{\phi}^*$ and $\partial\bar{\varepsilon}_F/\partial e$, which the model computes, the measurement needs no assumption on the distribution of shock sizes: whatever the distribution, the density at the margin cancels in the ratio of the two responses. The two responses identify the combined share $\tilde{\phi}$, not the probabilities ϕ_+ and ϕ_- separately.

Section 4 measures both responses, and Section 5 uses them to calibrate the model under an exponential distribution.

4 Empirical Evidence

This section measures the two responses that equation (26) combines and tests the predictions of Section 3.2. After describing the data (Section 4.1) and the variation in benefits (Section 4.2), I measure the response of temporary layoffs to the benefit (Section 4.3) and to experience rating (Section 4.4). Section 4.5 then asks whether firms deciding alone could produce the positive response to the benefit.

4.1 Temporary Layoffs in the Data

In the data, a worker on temporary layoff is an unemployed worker who has been given a date to return to work or expects to return to work within six months.[7] This is the definition of the Bureau of Labor Statistics (BLS), which is used in the design and publication of the Current Population Survey (CPS), and the literature follows it (e.g., Fujita and Moscarini 2017; Gallant et al. 2020; Gertler et al. 2026; Hall and Kudlyak 2022). I count all other unemployed workers as on permanent layoff, and I measure the permanent separations of the model by moves from employment into permanent layoff.[8]

Temporary layoffs are well documented in the studies cited above, and I summarize only the facts that this paper uses; Section 5 sets the transition rates of the model from several of them. I focus on private-sector workers aged 18 to 64 over 1994–2019, and I describe the construction of the CPS sample and the stock and flow of temporary layoffs in it in Appendix A.1.

I also use the Survey of Income and Program Participation (SIPP) in this paper, which allows me to follow workers month by month and to observe whether a worker returns to the previous employer. I construct the SIPP panel following Gertler et al. (2026), and I describe the construction in Appendix A.3. Two facts from the SIPP are useful for what follows. First, temporary layoffs are short: 90% of workers on temporary layoff are reemployed within six months, and 96% within

[7] See https://www.bls.gov/cps/definitions.htm#reasons

[8] Permanent layoff does not mean that workers are permanently unemployed. Besides permanent job losers, it includes workers whose temporary jobs ended, workers who quit, and workers who reentered the labor force.

twelve months. Second, most of them return to the previous employer: among workers reemployed within twelve months, 74% return to the employer they worked for before the layoff.[9] The remaining 26% move to a different employer, which is the move to a new employer at rate s in the model (Section 2.1).

4.2 UI Benefits and the Statutory Benefit Schedule

The first response that the model needs is the response of temporary layoffs to the benefit. UI in the U.S. is a joint federal-state program administered by the states. Although basic rules are set by the federal government, each state sets its own benefit schedule and changes it at its own times, and these changes are the variation I use (Appendix B).

The generosity of UI benefits depends on two variables: the weekly benefit amount (WBA) and the potential benefit duration (PBD). Each state sets a statutory schedule that maps a worker's previous earnings into the WBA. The schedule has three parts: a replacement coefficient, a minimum WBA, and a maximum WBA. Letting w denote the weekly earnings of a worker in the base period, the WBA in state s at time t is

$$b_{st}(w) = \min\left\{\max\left\{c_{st}\, w,\ \underline{b}_{st}\right\},\ \overline{b}_{st}\right\}, \tag{27}$$

where c_{st} is the replacement coefficient, $\underline{b}_{st}$ is the minimum WBA, and $\overline{b}_{st}$ is the maximum WBA. All three parameters differ across states, and each of them changes over time at different points in time. Since most temporary layoffs end within a few months (Section 4.1), I measure the generosity of UI by the WBA, which corresponds to the flow benefit b of the model.[10]

An important feature of the schedule (27) is that a change in a statutory parameter does not affect all workers in the same way. Figure 3 makes this concrete. Panel (a) plots the schedule for four states in 2010, which differ in all three parameters. Panel (b) takes two reforms and plots what each did across the earnings distribution. Raising the maximum, as California did in 2002, leaves the bottom nine bins untouched and raises benefits by more than a third at the top. Lowering the replacement rate, as Kentucky did in 2012, does the opposite, cutting benefits by about 9% in the ten lowest bins and leaving the ten highest almost unchanged, since the maximum binds there.

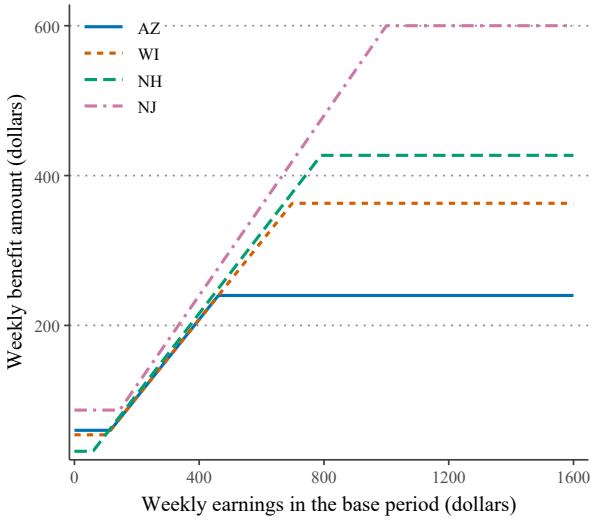
This is why the average benefit in a state is not a good treatment variable. It mixes these changes together and averages over workers who are affected and workers who are not, and a reform that moves benefits a great deal for the workers it reaches therefore appears as a small change in the

[9] Following Gertler et al. (2026), I count a return as a recall when the identifier of a job that pays the worker on return matches that of a job that paid the worker before the layoff (Appendix A.3).

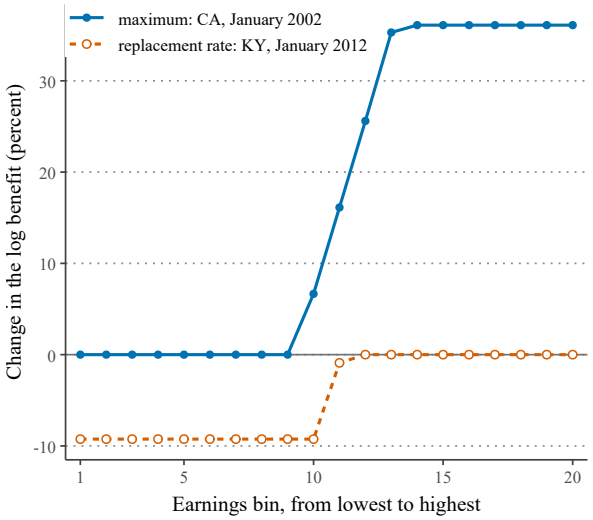
[10] The maximum PBD is 26 weeks in most states over most of the sample period, and in many states the PBD of a claimant also depends on earnings in the base period.

Figure 3: The Statutory Schedule and What a Change in It Reaches

(a) The Statutory Schedule in 2010



(b) Who a Change in a Parameter Reaches



Notes: Panel (a) plots the statutory schedule (27) in January 2010 for four states, chosen to span the range of maximum benefits. Panel (b) plots, for two reforms, the change in the log weekly benefit across the twenty fixed earnings bins described in the text. The two reforms are California in January 2002, which raised the maximum from $230 to $330, and Kentucky in January 2012, which lowered the replacement rate from 1.3078% to 1.1923% of base-period earnings. In each case the other two parameters, as coded in Appendix A.2, were unchanged.

state average.

I therefore construct a treatment variable that keeps the earnings dimension of the schedule. Following the simulated instrument approach of Currie and Gruber (1996a,b), I fix a national earnings distribution and apply each state's schedule to it. Specifically, I pool the real weekly earnings of private sector workers in the CPS over 1994–2019, in 2015 dollars of the consumer price index (OECD 2023), and divide them into twenty bins of equal size. Let $\bar{w}_k$ denote the median earnings of bin k, which is fixed across states and over time by construction. The simulated benefit for bin k in state s at time t is then $b_{st}(\bar{w}_k)$, obtained by converting $\bar{w}_k$ to current dollars, applying the statutory schedule (27), and expressing the result in 2015 dollars. Since the earnings distribution is held fixed, this variable moves only with the statutory parameters and, where the maximum or the minimum binds, with the price level, which erodes them between changes. It does not respond to changes in the composition of workers in the state, and it responds to the state's wages only where the maximum or the minimum is indexed to the state's average weekly wage (Appendix B).

I take the parameters of the statutory schedule from the semiannual publication *Significant Provisions of State Unemployment Insurance Laws* of the Department of Labor, for January and July of every year from 1994 to 2019 (U.S. Department of Labor 2019; Appendix A.2).

Do the statutory benefits reach workers on temporary layoff? The treatment variable is what the law says a worker in a given earnings bin is entitled to, not what the worker receives. In the SIPP, the benefits collected by all workers who separate into temporary layoff over their months on layoff add up to 46% of the statutory benefit for those months (Appendix A.3). Sections 5.1 and 5.2 use this ratio to scale the benefits of the model.

4.3 The Impact of UI Benefits on Separations

I estimate a linear probability model of the form

$$y_{iskm} = \beta \log b_{sm}(\bar{w}_k) + \psi_{sk} + \mu_{km} + \mathbf{x}'_{iskm}\gamma + \nu_{iskm}, \tag{28}$$

where the outcome y_{iskm} is either (i) an indicator for the transition from employment to temporary layoff or (ii) an indicator for the transition from employment to permanent separation, for individual i in state s in earnings bin k in year-month m. The treatment variable $\log b_{sm}(\bar{w}_k)$ is the log of the simulated benefit for the bin the worker belongs to. The term ψ_{sk} is a state-by-bin fixed effect and μ_{km} is a bin-by-year-month fixed effect. The vector $\mathbf{x}_{iskm}$ contains the full interaction of age, sex, race, and education. I multiply the outcome by 100 so that the estimated coefficient represents a percentage point impact, and I report the implied elasticity, which is the coefficient divided by the average outcome. Standard errors are clustered at the state level.

The two sets of fixed effects define the comparison the estimate is based on. The state-by-bin fixed effects absorb any permanent difference between, for example, low-earning workers in Michigan and low-earning workers in Texas. The bin-by-year-month fixed effects absorb anything that happens to a given part of the earnings distribution nationwide in a given month, such as a recession that hits low-wage workers first. What remains is variation within a state and within an earnings bin over time, net of what happens to the same earnings bin in the rest of the country. This variation comes from the timing and the shape of statutory changes: when a state raises its maximum WBA, benefits rise for high-earning workers in that state but not for low-earning workers in the same state, and not for high-earning workers elsewhere. The identifying assumption is that these changes are unrelated to other changes in the layoffs of the same state and earnings bin.

I estimate (28) separately in the CPS and in the SIPP. The two data sets are independent samples with different designs, and they measure temporary layoff differently. In the CPS, a worker on temporary layoff is unemployed and reports temporary layoff as the reason. In the SIPP, a worker on temporary layoff reports having a job but being absent from it on layoff without pay. I harmonize the samples otherwise: both are restricted to private for-profit wage and salary workers aged 18 to 64, outside agriculture, mining, and public administration. Appendix A.4 describes how the earnings bin is assigned in each data set.

Table 1: The Impact of UI Benefits on Separations to Unemployment

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Panel A. Transition from employment to temporary layoff							
Elasticity	0.499	0.433	0.421	0.538	0.516	0.931	0.435
	(0.267)	(0.261)	(0.394)	(0.276)	(0.278)	(0.418)	(0.247)
Cluster-robust p-value	0.067	0.103	0.290	0.057	0.069	0.030	0.084
Bootstrap p-value	0.056	0.089	0.315	0.015	0.016	0.038	0.099
Mean of the outcome	0.274	0.274	0.274	0.271	0.271	0.271	0.283
Panel B. Transition from employment to permanent separation							
Elasticity	-0.065	-0.075	-0.163	-0.141	-0.056	0.035	-0.108
	(0.131)	(0.135)	(0.220)	(0.116)	(0.109)	(0.189)	(0.161)
Cluster-robust p-value	0.621	0.582	0.463	0.228	0.609	0.852	0.504
Bootstrap p-value	0.621	0.591	0.480	0.313	0.616	0.858	0.499
Mean of the outcome	0.612	0.612	0.612	0.643	0.643	0.643	0.618
Sample	CPS	CPS	CPS	SIPP	SIPP	SIPP	CPS
State × bin fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bin × month fixed effects	Yes		Yes	Yes		Yes	Yes
Month fixed effects		Yes			Yes		
State × month fixed effects			Yes			Yes	
Reported earnings only							Yes
Observations	2,646,023	2,646,023	2,646,023	3,004,620	3,004,620	3,004,619	1,806,525

Notes: This table reports the estimates of β in the linear probability model (28) divided by the mean of the outcome, which gives the elasticity with respect to the statutory benefit. The treatment variable is the log of the simulated statutory weekly benefit amount for the earnings bin the worker belongs to. The outcome is an indicator multiplied by 100. Columns (1) to (3) use the CPS over 1994–2019 and columns (4) to (6) use the SIPP panels of 1996, 2001, 2004, and 2008. Column (7) repeats column (1) on CPS workers whose earnings in the fourth interview, which assign the earnings bin, were reported rather than allocated by the Census Bureau; the allocation flag is available from September 1995. All specifications include the full interaction of age, sex, race, and education. Survey weights are used. Cluster-robust standard errors, clustered at the state level, are in parentheses. The bootstrap p-value is from a restricted wild cluster bootstrap with Rademacher weights and 9,999 replications.

Table 1 reports the results. Columns (1) and (4) show the baseline specification in the CPS and in the SIPP. A 10% increase in UI benefits increases the transition from employment to temporary layoff by 5.0% in the CPS and by 5.4% in the SIPP. The two estimates are 0.499 and 0.538, with cluster-robust standard errors of 0.267 and 0.276. I emphasize that the two data sets are independent samples that measure temporary layoff differently, and their agreement makes it unlikely that the result comes from how one of the surveys records a temporary layoff.

Panel B shows that the same increase in UI benefits has no detectable impact on the transition from employment to permanent separation. The estimates are close to zero and statistically insignificant by either p-value in both data sets and in all specifications. As the model assumes when it treats permanent separations as exogenous (Sections 2.2 and 3.2), the temporary layoffs that the benefit induces do not come with more permanent separations.

Columns (3) and (6) add state-by-year-month fixed effects, so that the estimate uses only the

comparison across earnings bins within a state and month, which Section 4.5 discusses. Appendix A.6 shows that future changes in the benefit do not predict the transition into temporary layoff and reports further checks.

Inference. There are 51 state clusters, but they differ greatly in size, and the identifying variation, the variation in the benefit that remains after the fixed effects, is concentrated in a few of them. Tests based on the cluster-robust standard errors could then over-reject, and I therefore also report a bootstrap p-value, from a restricted wild cluster bootstrap with Rademacher weights (Cameron et al. 2008; Roodman et al. 2019). In the baseline specification, the bootstrap p-values are 0.056 in the CPS and 0.015 in the SIPP. The more conservative cluster jackknife gives p-values of 0.10 and 0.14 (Appendix A.5).

The positive response to the benefit is what joint decisions predict, and Section 4.5 asks whether firms deciding alone could produce it.

4.4 Experience Rating and Temporary Layoffs

The second response that the model needs is the response of temporary layoffs to experience rating. Experience rating charges a firm part of the benefits that its laid-off workers collect, through a higher payroll tax rate in later years. The experience-rating share is the present value of the additional taxes per dollar of charged benefits, times the share of benefits that are charged to the employer (Topel 1983; Card and Levine 1994), which Anderson and Meyer (2000) call the marginal tax cost.

Measuring the experience-rating share. I build the share from the state tax schedules that states report each year to the Department of Labor in the ETA 204 Experience Rating Report, 1998–2019 (U.S. Department of Labor 2026). For each interval of the benefit ratio or reserve ratio, the report gives the average tax rate assigned, the number of employers, their taxable payroll, and the benefits charged to them. These rows trace each state's schedule. The slope of the schedule between its minimum and maximum rates gives the tax cost per dollar of charged benefits, and employers on the minimum or maximum rate face none. To obtain variation within a state and year, I follow Card and Levine (1994) and place the representative employer of each NAICS sector on the state's schedule at the sector's steady-state cost rate, the benefits charged per dollar of taxable payroll. The cost rate combines the industry composition of UI claimants (ETA 203), the state's insured unemployment and average benefit (ETA 5159), and the sector's employment and taxable payroll in the Quarterly Census of Employment and Wages (QCEW; U.S. Bureau of Labor Statistics 2019). Appendix D.1 describes the construction.

The effective share. If this share measured how much a sector's taxes respond to its layoffs, a sector on the sloped part of its schedule would see its tax rate rise one for one with its cost rate. In the QCEW, the assigned tax rate of a sector rises by 0.25 for each point of its cost rate, and by 0.46 when the cost rate is instrumented with the layoff intensity of the sector in the CPS, which removes the measurement error of the cost rate. Taxes therefore respond much less to layoffs than the schedules imply, partly because many employers within a sector sit on the minimum or maximum rate. I define the effective share, the counterpart of e in the model, as the share that reproduces the response of the tax rate, and I measure the response of temporary layoffs per unit of this effective share.

Design. To use only variation that comes from changes in the schedules, I hold each sector's position on the schedule at its average over the sample and let the share move with the state's schedule alone. I run two regressions on the same state $\times$ sector $\times$ year cells, with the same employment weights and the same state $\times$ year, sector $\times$ year, and state $\times$ sector effects: the monthly transition from employment to temporary layoff in the CPS on the share, and the sector's tax rate on its cost rate interacted with the share, with the cost rate instrumented by the CPS layoff intensity. The ratio of the two coefficients is the response of temporary layoffs per unit of the effective share. An error that scales the share, such as the gap between the measured and the effective share, enters both regressions in the same way and cancels in the ratio.

Results. Table 2 reports the results, with the baseline specification in column (1). Panel A shows that the transition from employment to temporary layoff falls by 0.43 log points per unit of the measured share. Panel B shows that the sensitivity of the tax rate to the cost rate rises with the measured share at 0.18 times the rate that the schedules imply, and the effective share therefore averages 0.12, about a fifth of the share implied by the schedules. Panel C combines the two: per unit of the effective share, experience rating lowers the transition from employment to temporary layoff by 2.31 log points, with a 90% bootstrap interval from -6.11 to -0.35. Experience rating has no significant effect on permanent separations, as the model assumes: the response is -0.26 log points, with an interval from -0.99 to 0.62. The other columns take pieces of the baseline specification away, and the response is negative in every column. Appendix D.2 reports the first stage on the tax side and regressions on individual workers in the CPS.

4.5 Alternative Explanations

By Proposition 1, a firm deciding alone raises temporary layoffs when the benefit rises only if the expected loss of the match falls by more than the experience-rated charge. Unless the fall in the

Table 2: Experience Rating and Temporary Layoffs

	(1)	(2)	(3)	(4)
Panel A. Transitions, per unit of the measured share				
Employment to temporary layoff	−0.43	−0.36	−0.30	−0.54
	(0.24)	(0.35)	(0.15)	(0.28)
Employment to permanent separation	−0.05	−0.06	−0.04	−0.13
	(0.08)	(0.08)	(0.06)	(0.10)
Panel B. Tax rate				
Cost rate × measured share	0.23	0.27	0.27	0.16
	(0.08)	(0.16)	(0.12)	(0.11)
Cost rate	0.02	0.22	0.24	0.32
	(0.08)	(0.13)	(0.10)	(0.15)
Scale of the effective share, κ	0.18	0.22	0.22	0.13
Mean effective share	0.12	0.15	0.15	0.10
Panel C. Per unit of the effective share				
Employment to temporary layoff	−2.31	−1.63	−1.37	−4.10
	[−6.11, −0.35]	[−11.77, 19.94]	[−10.67, −0.36]	[−14.93, −0.47]
Employment to permanent separation	−0.26	−0.29	−0.18	−1.01
	[−0.99, 0.62]	[−1.33, 1.21]	[−1.41, 1.29]	[−3.16, 0.34]
Share from schedule changes only	Yes	Yes		
State × year, sector × year effects	Yes	Yes	Yes	Yes
State × sector effects	Yes			
Benefit-ratio states only				Yes
Cost rate in the tax regression	IV	IV	IV	IV
Cells	12,625	12,625	12,625	4,569

Notes: Each column runs two regressions on the same state × NAICS sector × rate-year cells, 2002–2019, with at least 30 CPS observations, weighted by private employment in the QCEW. Panel A regresses the monthly CPS transition of the cell on the measured experience-rating share and divides the coefficient by the mean transition. Panel B regresses the sector's assigned tax rate in the QCEW on its cost rate and on the cost rate times the measured share, with the cost rate and its product instrumented by the CPS layoff intensity of the sector in the three years before and its product with the share. κ is the coefficient on the product relative to the value of 1.24 that the schedules imply, one over the average charged share times the present value of a dollar of taxes paid in equal parts over the next three years. Panel C divides Panel A by κ. Standard errors in parentheses and 90% intervals in brackets are from 199 draws of a bootstrap that resamples states in both regressions jointly.

firm's surplus offsets the whole charge, this requires laid-off workers to move to new employers less often. A higher benefit could do so within the match, by lowering the search of laid-off workers, or through the labor market, and it could also raise the wage at which new matches form (Section 3.1.4). I examine the three channels in turn. The evidence on search also bears on the calibration of Section 5, which assumes that the rate of moving to a new employer does not respond to the benefit.

Table 3: Where Temporary Layoff Spells End

	(1)	(2)	(3)
Months after the separation	3	6	12
Panel A. Reemployed			
Log statutory benefit	0.009	-0.040	-0.092
	(0.171)	(0.049)	(0.054)
Cluster-robust p-value	0.959	0.424	0.096
Bootstrap p-value	0.959	0.405	0.108
Mean of the outcome	64.9	89.7	95.6
Panel B. Reemployed at a different employer			
Log statutory benefit	-0.253	-0.346	-0.459
	(0.714)	(0.521)	(0.493)
Cluster-robust p-value	0.725	0.509	0.356
Bootstrap p-value	0.730	0.505	0.361
Mean of the outcome	12.6	21.2	24.9
State × bin fixed effects	Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes
Spells	7,620	6,613	5,017

Notes: This table reports estimates of the effect of UI benefits on the outcome of a temporary layoff spell, divided by the mean of the outcome, which gives elasticities with respect to the statutory benefit. The sample is separations into temporary layoff in the SIPP panels of 1996, 2001, 2004, and 2008, for which every month up to the horizon is observed. The outcome in Panel A is an indicator for being reemployed within the horizon, and in Panel B an indicator for being reemployed at a different employer within the horizon, both multiplied by 100. The treatment variable is the log of the simulated statutory weekly benefit amount for the worker's earnings bin. Cluster-robust standard errors, clustered at the state level, are in parentheses. The bootstrap p-value is from a restricted wild cluster bootstrap with Rademacher weights and 9,999 replications.

Where temporary layoffs end. If the benefit works through retention, more generous benefits should reduce the rate at which laid-off workers leave for a different employer. I use the SIPP to look at where a spell ends: for each separation into temporary layoff I follow the worker for three, six, and twelve months and record whether the worker is reemployed and whether the worker is reemployed at a different employer.[11] Panel A of Table 3 shows that UI benefits do not change whether a worker is reemployed: the elasticities are close to zero at all three horizons, and at six months the standard error of the elasticity is 0.05. Panel B shows that the elasticity of moving to a different employer is −0.25 at three months, −0.35 at six months, and −0.46 at twelve months. The point estimates have the sign that a search response would produce, but none of them is statistically distinguishable from zero, with bootstrap p-values between 0.36 and 0.73. These estimates are too

[11] I use month rather than bin-by-month fixed effects, which would introduce about 2,000 to 2,600 cells in a sample of this size. The job identifier that distinguishes a recall from a new employer is preserved for workers who separate into temporary layoff (Appendix A.3).

Table 4: Job Search of Workers on Temporary Layoff

Panel A. Share who looked for work in the last four weeks (%)					
Weeks on layoff	All	< 5	5–13	14–26	> 26
Share looking	42.6	36.5	47.1	52.5	57.1
Observations	100,703	52,298	30,580	12,449	5,376
Panel B. Elasticity of the share looking with respect to the statutory benefit					
	(1)	(2)	(3)	(4)	(5)
Log statutory benefit	-0.056	-0.262	-0.103	0.037	0.173
	(0.187)	(0.255)	(0.256)	(0.314)	(0.489)
Cluster-robust p-value	0.764	0.310	0.689	0.906	0.725
Bootstrap p-value	0.754	0.283	0.715	0.906	0.674
Mean of the outcome	42.6	38.6	38.6	38.8	39.7
Sample	All interviews	Interviews 5–8	Interviews 5–8	Interviews 5–8	Interviews 5–8
Treatment	State index	State index	Own bin	Own bin	Own bin
State fixed effects	Yes	Yes			
Bin fixed effects		Yes			
State × bin fixed effects			Yes	Yes	Yes
Month fixed effects	Yes	Yes	Yes		
Bin × month fixed effects				Yes	Yes
State × month fixed effects					Yes
Observations	99,121	19,449	19,437	18,441	12,602

Notes: The sample is the stock of workers on temporary layoff in the CPS panel of Section 4.3, 1994–2019, ages 18–64. The outcome is an indicator for having looked for work in the last four weeks, multiplied by 100. Panel A reports its weighted mean, overall and by the number of weeks on layoff. Panel B reports the coefficient on the log statutory benefit divided by the mean of the outcome. Column (1) uses the state-level statutory index, the simulated benefit averaged over the fixed national earnings distribution, on every interview of every worker on layoff. Columns (2) to (5) use the stock observed in interviews 5 to 8; column (2) uses the state-level index, and columns (3) to (5) use the simulated benefit for the worker's own earnings bin, taken from the worker's earnings in interview 4. Demographic controls are included throughout. Cluster-robust standard errors, clustered at the state level, are in parentheses. The bootstrap p-value is from a restricted wild cluster bootstrap with Rademacher weights and 9,999 replications.

imprecise to tell whether benefits move the split between recalls and new employers.

Job search during temporary layoff. If the benefit works through retention, it should also reduce the share of laid-off workers who search at all. In most states, workers on temporary layoff are exempt from the job search requirements of UI as long as they expect to be recalled soon, unlike permanently laid-off workers, and whether they search is therefore their own choice.[12] Since 1994, a worker in the CPS who reports being on layoff is asked: even though you expect to be called back to work, have you been looking for work during the last four weeks? Panel A of Table 4 shows that 43% of the workers on temporary layoff in my CPS sample looked for work in the last four weeks: most of them wait for the recall, and a substantial minority search while they wait. Panel B asks

[12] See https://oui.doleta.gov/unemploy/pdf/uilawcompar/2023/nonmonetary.pdf

whether the benefit changes the share who search. Column (1) uses the state-level statutory index, the simulated benefit averaged over the fixed national earnings distribution, with state and month fixed effects, and it keeps every interview of every worker on layoff. The elasticity in column (1) is −0.06 with a standard error of 0.19, and the lower end of its 95% interval is −0.42 with the cluster-robust standard error and −0.62 with the restricted wild cluster bootstrap. This estimate detects no response of search participation to the benefit.

The labor market. A higher benefit could also lower vacancies, and with them the rate at which laid-off workers move to new employers, which would make a temporary layoff less risky for the firm even if laid-off workers searched as before. Columns (3) and (6) of Table 1 compare earnings bins within a state and month, which removes the response of a market in which all the workers of a state search, and the estimates are 0.42 in the CPS and 0.93 in the SIPP. In Table 3, the split between recalls and new employers does not change significantly. The evidence on the market effects of unemployment insurance also points to a market in which more generous benefits raise, rather than lower, the job-finding rate of other searchers (Marinescu 2017; Lalive et al. 2015).

The wages of new matches. A higher benefit could also raise the wage at which new matches form, which would lower the firm's threshold in those matches. Jäger et al. (2020) find that wages rise by less than one cent per dollar of benefit and reject sensitivities above three cents. In the calibrated model, unilateral decisions would raise temporary layoffs in a new match if its wage rose by more than about half a cent per dollar of benefit (Section 5.2). This channel therefore cannot be ruled out, but it works only in matches formed after the benefit changes and in existing matches as their wages are renegotiated (Section 6).

Summary. Taken together, the benefit changes the layoff without a visible change in what the worker does during it: reemployment is unchanged, the estimated shift toward recalls is imprecise, and search participation shows no detectable response. The data measure whether laid-off workers search, not how intensely they search or which offers they accept. In terms of the model of Section 2, the results limit the retention channel if the rate of moving to a new employer is proportional to search participation, and Section 5.2 asks how large a fall in search a firm deciding alone would need.

5 How Coasean Are Temporary Layoffs?

Section 4 finds that temporary layoffs rise with the benefit while the share of laid-off workers who search does not detectably fall, a pattern that, by Proposition 1, is hard to reconcile with decisions

made by firms alone. To go further, three objects are needed that the reduced-form responses do not reveal by themselves. First, the cutoff $\tilde{\phi}^*$ and the shift $\partial\bar{\varepsilon}_F/\partial e$ in equation (26) depend on how the benefit and experience rating move the firm's expected loss of the match, which involves the firm's value of a match on layoff and the rates at which layoffs end. Second, the data measure monthly transitions rather than the probability that a shock leads to a layoff. Third, the consequences in Section 6 depend on how a change in the wage moves the firm's values, which the two responses do not reveal. This section therefore calibrates the model of Section 2 to the observed transition rates and to the two responses measured in Section 4, and reads off the share of joint decisions. The calibration adds no identifying information beyond the assumptions of the model: it turns equation (26) into a number and extends it beyond the margin.

5.1 Calibration

Table 5 lists the parameters. Those in the upper panel are set outside the model, except the experience-rating share, which is the effective share measured in Section 4.4. Those in the middle panel are set from the data without the model, and those in the lower panel are chosen jointly to match the targets exactly.

Units. I take one unit of time to be a month, the frequency at which the data measure transitions, and I convert monthly transition probabilities into rates assuming that each rate is constant within the month. I normalize the wage of each match to one.

Transition rates. The transition rates come from the CPS sample of Section 4.1, workers aged 18 to 64 who stay in the labor force. A worker moves from employment to permanent separation with probability 0.010 per month, and a jobless worker finds a job within a month with probability 0.27. A worker on temporary layoff returns to employment within a month with probability 0.516, and I split this flow with the share of returns to the previous employer in the SIPP, 74.0% (Section 4.1): the worker is recalled within a month with probability 0.382 and moves to a new employer with probability 0.134. These targets give a permanent separation rate of $\delta = 0.010$, a job-finding rate of $f = 0.312$, a recall rate of $\lambda_r = 0.541$, and a rate of moving to a new employer of $s = 0.190$. The model leaves out the 15% of workers on temporary layoff who are classified as on permanent layoff in the following month, since in the model a match on layoff ends only through a recall, a move to a new employer, or a permanent separation. The calibration holds s fixed when the benefit changes, $\eta_{s,b} = 0$, which the evidence on search in Section 4.5 cannot reject. If a higher benefit lowered s, the firm's own decisions would account for more of the measured elasticity, and the share of joint decisions would be smaller: 0.07 with $\eta_{s,b} = 0.2$ and 0.025 with $\eta_{s,b} = 0.42$, the cluster-robust end of the 95% interval in column (1) of Table 4.

Table 5: Parameters of the Calibrated Model

Parameter	Description	Value	Source or target
Set outside the model			
r	Discount rate	0.0040	Annual discount factor 0.953
w	Wage	1	Normalization
b_k	Benefit in bin k	0.16–0.67	Statutory schedules, Sec. 4.2
e	Experience-rating share	0.125	Effective share, Sec. 4.4
ϕ_-	Prob. of agreement to keep working	0	Davis and Krolikowski (2025)
Set from the data without the model			
δ	Permanent separation rate	0.0102	CPS E→PL 0.010
f	Job-finding rate	0.312	CPS PL→E 0.27
λ_r	Rate at which shocks end	0.541	Recall 0.38 (CPS, SIPP)
s	Rate of moving to new employer	0.190	New employer 0.13 (CPS, SIPP)
ℓ	Value of non-work time	0.478	Negative-gain share of TL 52%
Calibrated jointly			
ϕ_+	Prob. of agreement to lay off	0.17	Benefit elasticity 0.50
μ	Mean shock size	0.080	Response to ER −2.31
$\lambda_{d,k}$	Rate at which shocks arrive in bin k	0.120–0.265	E→TL by bin, 0.41% overall
y	Output in normal times	1.0420	Value of a new match 1.30 months of output

Notes: Rates are per month. The wage, the job-finding rate, and the rate of moving to a new employer are those of a labor market held fixed (Section 2). The benefit and the rate at which shocks arrive differ across the 20 earnings bins of Section 4.2; the table reports their ranges. ER: experience rating; TL: temporary layoff; PL: permanent layoff.

Unemployment insurance. Matches differ in the benefit relative to the wage. I group workers into the 20 earnings bins of Section 4.2 and give the matches in bin k the benefit b_k, the statutory replacement rate of the bin, averaged over states and years with the weights of the workers in the bin. The replacement rate is 0.67 in the lowest bin, about 0.52 in the bins up to the middle of the distribution, and it falls from there, where the maximum benefit starts to bind, to 0.16 in the highest bin. I use the statutory rate, as if every worker on temporary layoff collected the benefit, although workers collect 46% of it (Section 4.2). If the benefit collected moves in proportion to the statutory benefit, this changes the scale of the benefit but not its elasticity, and since experience rating charges the benefits collected, a common scale of the benefit and the charge cancels from equation (26): recalibrating the model with benefits scaled by 0.46 gives the same share of joint decisions (Appendix C). The scale matters for the search response that a firm deciding alone would need (Section 5.2) and for the level of the response to the real wage (Section 6). The experience-rating share is the effective share measured in Section 4.4, $e = 0.125$.

Value of non-work time. The value of non-work time ℓ is common to all matches, and together with the replacement rates it determines which matches are negative-gain matches: those in the bins with $b_k + \ell < 1$. No direct estimate of ℓ is available, and I choose it so that negative-gain matches account for about half of temporary layoffs, the lower bound implied by the survey of Davis and Krolikowski (2025), in which 54.5% of the workers on temporary layoff would have accepted a temporary pay cut of 5% to avoid the layoff (Appendix C). This gives $\ell = 0.478$, and Section 5.2 reports the results when the share is near 32% or near 64%.

Agreements. The probability of an agreement to keep working is $\phi_- = 0$. In the survey of Davis and Krolikowski (2025), a cut in pay, benefits, or hours was discussed as an alternative to only 4.2% of temporary layoffs (Section 2.2), which suggests that such agreements are rare. The survey covers workers who were laid off, however, and it does not see agreements that kept a worker at work. Section 5.2 therefore reports the results with $\phi_- = 0.1$ and 0.25. The probability of an agreement to lay off, ϕ_+, is calibrated below.

Shocks. To calibrate the model, I assume that shock sizes are exponentially distributed with mean μ. The density at the margin in equation (15) is then $\bar{h}_k = 1/\mu$ in matches of either type and wherever the thresholds are, as long as they are positive, so that $\bar{h} = 1/\mu$ across all matches. Since $g(\bar{\varepsilon}_C)/g(\bar{\varepsilon}_F) = \exp(\Delta_W/\mu)$, the share of joint decisions at the margin in a positive-gain match with the worker's gain Δ_W is

$$\tilde{\phi}_+ = \frac{\phi_+}{\phi_+ + (1 - \phi_+)\exp(-\Delta_W/\mu)}, \tag{29}$$

and it is zero in a negative-gain match when $\phi_- = 0$. By equation (24), the share across all matches is then the average of these shares weighted by each bin's share of temporary layoffs. The rate at which shocks arrive in bin k, $\lambda_{d,k}$, matches the bin's monthly probability of moving from employment to temporary layoff (Appendix C).

The two responses. The probability of an agreement ϕ_+ and the mean size of shocks μ match the two responses. Both hold the labor market of a state fixed, as long as the earnings bins and sectors of a state share its labor market. The response to experience rating compares sectors within a state and year, and the benefit elasticity in the CPS is about the same when it compares earnings bins within a state and month (Section 4.5). I therefore match them with the responses of the model across all bins, in which the wage and the labor market are fixed: a benefit elasticity of the transition from employment to temporary layoff of 0.50, the CPS estimate in column (1) of Table 1, and a response of −2.31 log points per unit of the effective experience-rating share (Section 4.4). Since the benefit differs across bins, both policies shift the thresholds of a bin in proportion to its benefit,

and the two responses weight the bins by their benefits rather than as in equation (24); I therefore compute both responses in the model rather than apply equation (26) directly.

Output. Output y sets the firm's value of a new match, which I set to 1.30 months of output on average over bins, the expected cost of a hire implied by the vacancy cost in Blanco et al. (2026) at the job-finding rate that their calibration targets; with exponential shocks, the share of joint decisions at the margin does not depend on this value as long as both thresholds are positive (Appendix C).

5.2 The Share of Joint Decisions

Table 6 reports the calibrated economy and two benchmarks with the same distribution of shocks. The calibration gives $\phi_+ = 0.17$. The share of joint decisions at the margin is $\tilde{\phi} = 0.12$ across all matches and 0.24 in positive-gain matches, the matches in which an agreement to lay off can move the decision. About 12% of the layoff decisions at the margin weigh the worker's payoff, and about a quarter among the matches whose workers gain from a layoff.

The two responses identify different objects, as equation (25) implies. The response to experience rating is almost the same whatever the share of joint decisions, from −2.29 when the firm decides alone to −2.17 when every decision is joint, and it pins down the density of shocks at the margin, $1/\mu = 12.5$. Given this density, the benefit elasticity rises steeply with the share of joint decisions, from −0.29 when the firm decides alone to 5.81 when every decision is joint, and the measured 0.50 lies in between. Applying equation (26) to each draw of the bootstrap of Table 2, with the model's terms in the equation at their calibrated values, gives a 90% interval for the share of joint decisions of $[0.06, 0.34]$. It conditions on a negative response to experience rating, as the model requires, which leaves out 2.5% of the draws, and it does not include the sampling error of the benefit elasticity. The benefit elasticity also depends on the sample. On the sample of the experience-rating regressions, the CPS over 2002–2019 with sector-by-year effects, it is 0.75 (Table D.4), and matching it instead of 0.50 gives a share of joint decisions of 0.15.

The value of non-work time and agreements to keep working. Table 7 reports the share of joint decisions when negative-gain matches account for a different share of temporary layoffs and when agreements to keep working are possible. Across these calibrations, the share across all matches lies between 0.11 and 0.12, and the share in positive-gain matches between 0.09 and 0.31. When more matches are negative-gain, more of the margin is decided by firms alone, and the measured elasticity requires agreements to lay off to be more common in the remaining matches. Agreements to keep working add joint decisions in negative-gain matches, and fewer agreements to lay off are

Table 6: The Share of Joint Decisions

	Firm alone $\phi_{\pm} = 0$	Calibrated $\phi_{+} = 0.17, \phi_{-} = 0$	Joint $\phi_{\pm} = 1$
Share of joint decisions at the margin, $\tilde{\phi}$	0.00	0.12	1.00
in positive-gain matches	0.00	0.24	1.00
Benefit elasticity of E→TL	−0.29	0.50	5.81
Response to experience rating (per unit of e)	−2.29	−2.31	−2.17

Notes: The calibrated economy matches a benefit elasticity of 0.50 and a response of −2.31 log points of the transition from employment to temporary layoff per unit of the effective experience-rating share, both across all earnings bins, in matches with a given wage, and with the labor market fixed. The second row is the share of joint decisions at the margin among positive-gain matches. The benchmarks keep the calibrated distribution of shocks and output and refit the rates at which shocks arrive to the level of temporary layoffs by bin.

then needed. The share across all matches thus depends little on where joint decisions occur: the two responses measure it without separating the two types of match (Section 3.2).

Table 7: The Share of Joint Decisions under Other Calibrations

	Negative-gain share of temporary layoffs	ℓ	ϕ_{+}	$\tilde{\phi}$	$\tilde{\phi}$ in positive-gain matches
Baseline	0.52	0.478	0.17	0.12	0.24
Negative-gain share near 32%	0.32	0.516	0.08	0.11	0.16
Negative-gain share near 64%	0.63	0.475	0.22	0.12	0.31
$\phi_{-} = 0.1$	0.52	0.478	0.12	0.11	0.18
$\phi_{-} = 0.25$	0.52	0.478	0.05	0.11	0.09

Notes: Each row recalibrates ϕ_{+}, μ, the rates at which shocks arrive, and output to the same targets as the baseline. The share of temporary layoffs in negative-gain matches moves in steps of one earnings bin. The baseline uses the step closest to 54.5%, the share of workers on temporary layoff in Davis and Krolikowski (2025) who would have accepted a 5% pay cut to avoid the layoff, and the next two rows use the steps closest to 32% and 64%.

Unilateral decisions. A firm that decides alone lowers temporary layoffs when the benefit rises, with an elasticity of −0.29: the experience-rated charge outweighs the fall in the expected loss of the match. In the calibrated model, the charge raises the firm's threshold by $e = 0.125$ per unit of benefit and the fall in the firm's surplus lowers it by $R_b = 0.077$ on average, which puts the cutoff of Proposition 2 at $\tilde{\phi}^* = 0.048$. Even a zero response would require the rate of moving to a new employer to fall by 0.20% for each 1% increase in the benefit, and reaching the measured elasticity would require 0.55%. If the rate of moving to a new employer is proportional to the share of laid-off workers who search, 0.55% lies outside the cluster-robust 95% interval of column (1) of Table 4 but inside its bootstrap interval (Section 4.5). The required fall scales with the charge

relative to the expected loss of the match, $eb/(sJ_T)$ in condition (20), and recalibrating the model with benefits scaled by 0.46 lowers it to 0.45%, beyond the cluster-robust end of column (1) but inside its bootstrap interval (Appendix C).

The wages of new matches. In the calibrated model, a lasting rise in the wage of a match lowers the firm's threshold by 9.7 times the rise in the wage, mostly through the firm's value of a match on layoff, the sensitivity that also drives Section 6. Since a higher benefit raises the firm's threshold by $e - R_b = 0.048$, unilateral decisions would raise temporary layoffs in a new match if its wage rose by more than 0.5 cents per dollar of benefit.

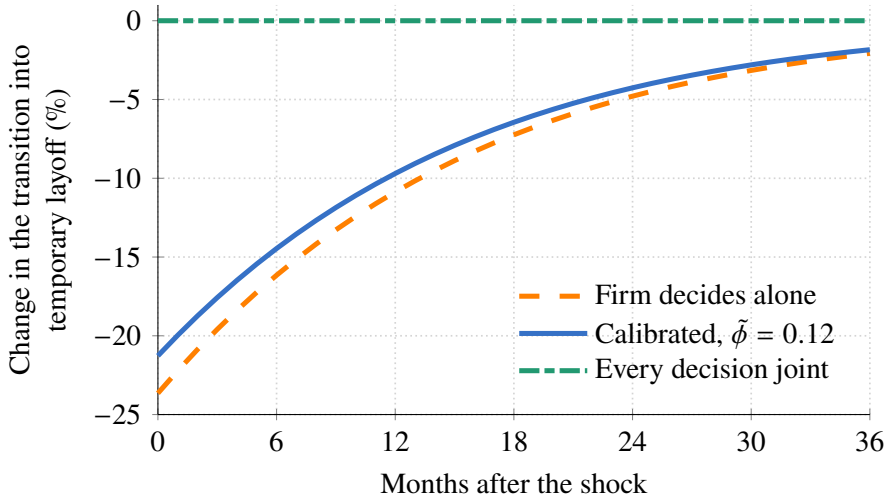
6 Consequences of Partially Coasean Temporary Layoffs

The share of joint decisions governs how rigid wages transmit shocks into temporary layoffs. As Barro (1977) argued, a rigid wage need not distort employment when the firm and the worker can agree on the allocation, and in the model only the decisions that the firm makes alone depend on the wage. This section shows what the measured share implies for this exposure.

A rise in the price level. Suppose that nominal wages are fixed and that the price level rises unexpectedly by 1%. The real wage of existing matches then falls about 1% below the wage at which new matches form. As in Blanco et al. (2026), the wage of a match is renegotiated at a monthly rate of 0.071, which matches the monthly probability of a base wage change among hourly workers who stay in their jobs, 6.9%, in Grigsby et al. (2021), and a renegotiation brings it to the wage of new matches, so that the deviation decays with a half-life of about ten months. As in the measurement, I hold the labor market, the wage of new matches, and the real benefit fixed: the experiment isolates the rigidity of existing wages rather than the full effect of inflation. A shock to productivity would instead move the firm's and the pair's thresholds alike, as experience rating does, and its effect would not depend on the share. A change in the real wage separates the two decisions.

The two decisions. A lower wage raises the flow that the firm earns while producing and its value of a match on layoff, and the firm therefore lays off only after larger shocks. For the pair, the wage only divides the value of the match between its two parties. When the wage of a match differs from the wage of new matches, a move to a new employer changes the worker's payoff as well as the firm's ($\tilde{W}_0 \neq W_T$), and the pair, which counts both, lays off after almost the same shocks as before. Treating the pair's threshold as fixed, the wage shifts only the firm's threshold in equation (24), and

Figure 4: Temporary Layoffs after a 1% Rise in the Price Level



Notes: Response of the monthly transition from employment to temporary layoff in existing matches when a 1% rise in the price level, with nominal wages fixed, lowers the real wage of existing matches by 1%, and wages are renegotiated to the wage of new matches at a monthly rate of 0.071. The labor market, the wage of new matches, and the real benefit are held fixed. The three economies share the calibrated distribution of shocks and differ in the probabilities of agreements. The responses are first order in the deviation and hold the pair's threshold fixed; to first order, staggered renegotiation acts as a common deviation that decays at the monthly rate of 0.071.

the response is the first term alone: only the part of the margin that the firm decides alone, $1 - \tilde{\phi}$, is exposed to the rigidity.

Results. Figure 4 shows the response of the transition from employment to temporary layoff. When the firm decides alone, the transition falls by 24% on impact. In the calibrated economy, it falls by 21%, and when every decision is joint, it does not move. In log points, the calibrated response is 0.89 times the response when the firm decides alone, close to $1 - \tilde{\phi} = 0.88$, and the ratio barely depends on how long the deviation lasts: it is 0.89 for a half-life of 12 months and for a lasting deviation.[13] More generally, as the probabilities of agreements vary from zero to one with the distribution of shocks fixed, the ratio stays close to $1 - \tilde{\phi}$: at shares of joint decisions of 0.18, 0.34, 0.50, and 0.68, it is 0.82, 0.66, 0.50, and 0.32. For this shock, the unilateral benchmark is therefore a close approximation: when nominal wages are rigid, the calibrated response to inflation and deflation is only about 11% smaller. The share is measured with error, and the ratio inherits it nearly one for one. The level of the response is large because the firm's margin over the wage while producing is thin, and it depends on how long the deviation lasts and on the scale of the benefit; I therefore emphasize the ratio.

[13]These responses hold the pair's threshold fixed. Solving for it, with $\tilde{W}_0$ in equation (5) equal to the value of waiting on layoff in a match at the wage of new matches, changes the ratio for a lasting deviation from 0.894 to 0.887.

What the measurement adds. Two features of the measurement drive this result. First, it measures a degree rather than choosing between two views. The calibrated economy lies close to the model in which the firm decides alone and far from the model in which every decision is joint, as in Feldstein (1976) and Baily (1977). Yet the same minority of joint decisions that barely changes the response to the real wage reverses the response to the benefit (Section 5). The difference lies in how each policy or shock moves the two thresholds. A change in the real wage moves only the firm's threshold, and the response falls in proportion to the share decided by the firm alone. A change in the benefit moves the two thresholds in opposite directions, and joint decisions respond far more strongly than the firm's own: the benefit elasticity is -0.29 when the firm decides alone and 5.81 when every decision is joint, so that a share of about 5% already reverses the sign. Experience rating, like a shock to productivity, moves both thresholds alike, and its effect barely depends on the share. Second, the wage is equally rigid in all three economies of Figure 4. What differs is whether the decisions weigh the worker's payoff, not whether the wage adjusts. A rigid wage therefore does not by itself expose layoffs to the rigidity, and the measured share tells how much of the margin is exposed.

7 Conclusion

This paper measures the extent to which temporary layoff decisions take workers' payoffs into account. Changes in U.S. unemployment insurance provide two complementary responses: higher benefits increase temporary layoffs, whereas stronger experience rating reduces them. The benefit response alone cannot reveal how common joint decisions are, because it also depends on how many matches lie near the layoff threshold. Experience rating provides information about this margin, allowing the two responses to identify the share of joint decisions in the model.

The finding that only 12% of marginal decisions are joint suggests that temporary layoffs are predominantly unilateral. Yet this does not mean that the unilateral model adequately predicts their response to every policy or shock. Even this minority of joint decisions is enough to reverse the predicted response to UI benefits: whereas unilateral decisions imply that higher benefits reduce temporary layoffs, joint decisions need to carry only about 5% of the margin for higher benefits to raise them, as the data show. In contrast, unilateral decisions provide a closer approximation to the transmission of wage rigidity. Following a decline in the real wage of existing matches, temporary layoffs respond about 89% as much as under purely unilateral decisions. Thus, whether unilateral decision-making is a useful approximation depends not only on how prevalent joint decisions are, but also on how a policy or shock affects the incentives of firms and workers.

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Online Appendix (Not for Publication)

A Data and Empirical Details

A.1 Current Population Survey

The Current Population Survey (CPS) is the monthly household survey that the Census Bureau conducts for the Bureau of Labor Statistics. It is a source of labor market statistics in the U.S. The CPS has a short panel structure. Each household in the CPS is interviewed each month for 4 months, then remains out of the sample for 8 months, and is then interviewed again for 4 more months. Although each household is interviewed (up to) 8 times, earnings information in the monthly CPS is asked only in the 4th and 8th interviews. Respondents in the 4th or 8th interviews are called the Outgoing Rotation Group (ORG).

I obtained the monthly samples of the CPS over 1994–2019 from IPUMS (Flood et al. 2023). The questions on job search during a layoff (Section 4.5) and the flags for allocated earnings (Section 4.3) come from a later extract of the same samples (Flood et al. 2025). The monthly CPS samples contain information on labor market status and other demographic variables.

Employment status. I use `EMPSTAT` to identify the employment status of each individual. I classify workers who are either "at work" or "has a job, not at work last week" as employed. I drop workers who are not in universe (those under age 15), members of the armed forces, and those not in the labor force. The remaining workers are classified as unemployed, and further as on temporary layoff or on permanent layoff.

Temporary layoff and permanent layoff. Following Fujita and Moscarini (2017), I identify workers on temporary layoff with the reason for unemployment, `WHYUNEMP`. `WHYUNEMP` can take the following six categorical values (excluding 0 for not-in-universe): 1. Job loser - on layoff, 2. Other job loser, 3. Temporary job ended, 4. Job leaver, 5. Re-entrant, 6. New entrant. `WHYUNEMP=1` is classified as temporary layoff, while `WHYUNEMP=2,3,4,5` are classified as permanent layoff. I drop workers in `WHYUNEMP=6` from the sample.

Worker flows. Although the CPS has a panel structure, there is no individual identifier in the original survey that allows linking across samples. IPUMS provides `CPSIDV` for linking individuals across different months, which allows me to look at labor market transitions for each worker. I link each interview to the interview of the following calendar month, including December to January.

Additional sample selection. `CLASSWKR` identifies whether a worker is self-employed, working in the private sector, working in the public sector, working in the armed forces, or working without pay in a family business or farm. For unemployed workers, the classification is according to the most recent job. I restrict the sample to workers who are working for a private for-profit employer or unemployed workers whose most recent job was with one. I similarly remove employed workers in or unemployed workers from the agricultural, mining, or public administration sectors. These restrictions apply to the month in which a transition starts. In the following month I record the state of every worker, whatever the class of worker and the industry, and a worker who leaves the labor force counts as making neither a temporary layoff nor a permanent separation, as in the SIPP.

Temporary layoffs in the sample. In the sample of private-sector workers aged 18 to 64 over 1994–2019, 12% of unemployed workers are on temporary layoff, while 29% of the inflow to unemployment is accounted for by temporary layoffs. The difference between the stock and the flow comes from the fact that workers on temporary layoff quickly get reemployed. Among workers who stay in the labor force, the monthly job finding rate of workers on permanent layoff is 27%, while that of workers on temporary layoff is 52%. Another 15% of workers on temporary layoff are classified as on permanent layoff in the following month, the loss of recall that Gertler et al. (2026) emphasize.

A.2 The Statutory Benefit Schedule, 1994–2019

The treatment variable in Section 4.3 requires the three parameters of the statutory schedule (27) for every state and every period. These parameters are published by the Department of Labor in *Significant Provisions of State Unemployment Insurance Laws*, which appears twice a year, in January and in July. I assign the parameters of each issue to the six months that begin with it, so that a change made between two issues takes effect in the data at the second. They are not available in machine-readable form, and I therefore build the schedule from the original publication for every issue from January 1994 to July 2019, which is 51 states by 52 issues.

Issues from 2000 onward. For these issues the published PDF files contain a text layer, from which the table can be read directly. I extract the position of every word on every page and locate the three columns of interest, "Computation of Weekly Benefit Amount," "Minimum," and "Maximum," by finding the column headers and assigning each word to the state whose row it belongs to. I then code the replacement coefficient by hand from the printed formula. This is necessary because the formula is stated in words and in a variety of forms: some states compute the weekly benefit as a fraction of the highest quarter of base period earnings, others as a fraction

of annual earnings, and some use a sliding scale. For a worker with steady earnings w per week, earnings in the highest quarter are $13w$ and earnings in the base period are $52w$, and each printed formula therefore maps into a single coefficient c_{st}. In some states the printed formula is a sliding scale, a bracket schedule, or another formula that is not proportional to earnings, and I approximate it by a single coefficient within its range; California's scale from 1/23 to 1/26 of high-quarter earnings, for example, becomes 1/25. In Utah from 2011 the formula subtracts five dollars, and in Pennsylvania from 2017 it adds about two dollars, which the simulated benefit includes. I record a formula as changed only when the printed text changes, and I verify each change against the previous and the following issue.

Issues from 1994 to 1999. For these issues the publication is available only as scanned images with no text layer. I therefore have each page transcribed twice, independently, together with a record of which characters the reader was unsure of. The two transcriptions agree on 99.8% of the maximum benefits, on 100% of the minimum benefits, and on 99.8% of the formulas. Where they disagree, or where a reader was unsure, I compare them with two independent sources: the state UI law data of Chetty (2008), which covers the minimum and maximum benefit for 41 states and the District of Columbia through the first half of 2000, and the dated benefit formulas in the same replication package. This resolves most of the flagged cells. For the cells that remain I use three internal consistency checks. First, in states that index it to the state average weekly wage, the maximum benefit moves smoothly. A value that matches neither the previous nor the following issue is suspect. Second, in states that print a range, the two numbers are tied to each other by the statutory dependants' allowance, and a misread digit breaks the ratio. Third, a formula that appears in no other issue of the same state is suspect, since a repeated misreading is unlikely. After these checks, seven cells required me to inspect the original image, and I did so.

Ranges and dependants' allowances. Several states print a range rather than a single maximum. In almost all of them the lower number is the maximum benefit for a worker with no dependants and the higher number includes the dependants' allowance. I use the lower number throughout, because neither the CPS nor the SIPP identifies the number of dependants a worker has. Minnesota, and Colorado from July 2007, are the exceptions: they have no dependants' allowance, and their printed ranges give the caps on two alternative computations, of which the law pays the higher. The publication is not consistent about how it prints Minnesota, however: it shows a single number until January 1999 and again from July 2001 to January 2006, and a range otherwise. Neither number gives a series free of this convention: taking the higher number where a range is printed would put a 22% jump in the series in 1999, and taking the lower number puts a rise of 37% in July 2001 and a fall of 32% in July 2006. I use the lower number for both states, the convention of Chetty (2008)

for Minnesota; excluding Minnesota raises the benefit elasticity from 0.50 to 0.55 in the CPS and from 0.54 to 0.58 in the SIPP.

A.3 The SIPP Panel

The SIPP is a set of panels, each following a sample of households for three to five years. Households are interviewed every four months, and each interview collects information for the preceding four months, called a wave. I use the panels beginning in 1996, 2001, 2004, and 2008 (U.S. Census Bureau 2013), and I construct the monthly panel following Gertler et al. (2026).

Labor force status. I measure labor force status in the second week of each month, using the recode `RWKESR2`, which is also the measurement used by Fujita and Moscarini (2017). A worker is employed if the recode indicates a job, either working or absent but not on layoff. A worker is on temporary layoff if the recode indicates having a job but being absent from it on layoff without pay. A worker is on permanent separation if the recode indicates no job while looking for work or on layoff. I exclude the panels beginning before 1996 because they classify many workers on temporary layoff as out of the labor force, as Fujita and Moscarini (2017) document. I also exclude type-Z observations, for which the Census Bureau imputes the entire record.

Sample. The flow sample consists of person-months in which the worker is employed, the main job is a private for-profit wage and salary job outside agriculture, mining, and public administration, the worker is aged 18 to 64, and the worker is observed in the following month. When a worker holds two jobs I treat the one with the larger monthly earnings as the main job. The SIPP groups the smallest states together in some panels, and those person-months are dropped since no state schedule can be assigned to them.

Base period earnings and the earnings bin. For a transition that starts in calendar quarter q, I take the base period to be quarters $q-5$ through $q-2$, which is the definition used by most states and follows Kuka (2020). Quarterly earnings are the sum of monthly earnings over the three months, counted only for quarters that are observed in full. I take the highest of the observed base period quarters, of which at least one must be observed in full, and divide by thirteen to obtain the weekly earnings that the statutory schedule applies to. The earnings bin is then assigned by cutting these weekly earnings, in 2015 dollars, at the twenty quantiles of their distribution in the pooled flow sample, and, as in Section 4.2, the simulated benefit applies each state's schedule to the median earnings of the bin. As in the CPS, the schedule and the price level are those of the month in which the transition starts.

Seam effects. Because each interview collects four months at once, transitions are reported disproportionately at the boundary between waves. In my flow sample 31% of transitions from employment to temporary layoff are reported at a seam, against the 25% that would be reported there if transitions were spread evenly across the four months of a wave. I include an indicator for the seam in all specifications.

Job identifiers and the limits of the data. The SIPP assigns each worker a number for each employer, which makes it possible to tell whether a worker returns to the employer held before a jobless spell. The identifier is lost when a worker who separates permanently is jobless for an entire wave, but it is preserved for workers who separate into temporary layoff, the spells I study (Fujita and Moscarini 2017). Among these workers, 81% of those who return to work within three months return to the previous employer, against 30% of those who take eight months or more. In the model, the rates of recall and of moving to a new employer are constant, and the share still falls with the length of the spell, since a match that ends during the layoff leaves only new employers. The survey also uses dependent interviewing, under which a worker's reported status rarely changes during a jobless spell. A worker who begins a spell expecting a recall and later gives up on it is therefore not observed switching status, which is why Section 4.5 measures where a spell ends rather than what happens during it.

UI receipt. Among workers separating into temporary layoff whom I observe for the following six months, 38% report receiving UI in those months, and the UI that all these workers report over the six months, counting the zeros of those who report none, adds up to 46% of the statutory benefit for their months on layoff, which run until reemployment or for six months (Section 4.2). Among those who receive UI, with state-by-bin and month fixed effects, a one log point increase in the statutory benefit for the worker's earnings bin raises the total amount received over six months by 0.24 log points, with a standard error of 0.37, so that these data cannot tell how closely the amount received follows the statutory benefit. If the amount received rose by less than the statutory benefit, the measured elasticity would understate the response to the amount received, and the share of joint decisions in Section 5.2 would be larger.

A.4 Assigning the Earnings Bin in the CPS

The CPS asks about earnings only in the fourth and eighth interviews, and the earnings bin therefore cannot be assigned to every worker in every month. This creates a problem that is easy to miss. If I assign the bin using earnings from whichever of the two interviews is available, then a worker who is on temporary layoff in the fourth interview has no earnings recorded and is dropped. The

bin is then assigned on the basis of the outcome. The size of the problem is large: among workers observed in the third or seventh interview, the transition rate to temporary layoff is 0.18% for those with recorded earnings and about 8% for those without.

I therefore assign the bin using earnings recorded in the fourth interview only, and I use it for the same worker's transitions out of the fifth, sixth, and seventh interviews. The earnings are then recorded nine to eleven months before the interview at which the transition starts. They are determined before the outcome and cannot be affected by it. This keeps about a third of the person-months that an assignment from either interview would use, and the remaining sample is the one used throughout Section 4.3.

I harmonize the CPS sample with the SIPP sample in two ways, neither of which is innocuous. The CPS sample is restricted to private for-profit wage and salary workers, outside agriculture, mining, and public administration, which the SIPP sample is not by construction, and I impose the same industry and class-of-worker restriction on the SIPP. The SIPP sample is restricted to workers aged 18 to 64, which the CPS is not, and I impose the same age restriction on the CPS. The age restriction matters: workers outside this range have transition rates to temporary layoff of 0.47% and 0.49%, 1.7 to 1.8 times the 0.27% of workers aged 18 to 64, and they are largely outside the UI system, since workers under 18 rarely have the base period earnings that qualify them and most separations of workers over 64 are retirements.

A.5 Inference

Table A.1 reports the baseline specification together with three measures of precision and with the features of the clusters that make them differ.

Panel A shows that the three measures do not agree. The cluster jackknife gives a larger standard error than the usual cluster-robust estimator, and the bootstrap gives a smaller p-value than either. Panel B shows why. There are 51 state clusters, but they are very unequal: the largest state contributes far more observations than the median one, and, more importantly, the share of the identifying variation contributed by the largest state is many times what it would be if the variation were spread evenly. The effective number of clusters, which summarizes this inequality, is 12.5 in the CPS and 8.3 in the SIPP. MacKinnon et al. (2023) show that inference based on the cluster-robust estimator with a t distribution can be unreliable in exactly this configuration, and they advise trying other methods when the methods disagree.

I therefore run placebo regressions in the spirit of their Section 3.5. In each replication I give every state another state's entire statutory schedule, holding the earnings bins, the calendar, the outcomes, and the fixed effects fixed. The reassigned schedule should have no effect, and it has the same serial and cross-sectional structure as the actual treatment, which a randomly generated

Table A.1: Inference and the Structure of the Clusters

	(1)	(2)
Sample	CPS	SIPP
Panel A. Estimate and three ways of measuring its precision		
Elasticity	0.499	0.538
Cluster-robust standard error (CV1)	(0.267)	(0.276)
Cluster jackknife standard error (CV3)	(0.298)	(0.354)
p-value, CV1	0.067	0.057
p-value, CV3	0.100	0.135
p-value, wild cluster bootstrap	0.056	0.015
Panel B. Why they differ: the clusters		
Number of state clusters	51	51
Effective number of clusters	12.5	8.3
Largest share of the identifying variation in one state	0.21	0.30
Median observations per state	38,767	42,422
Largest observations per state	208,187	291,986
Panel C. Leaving one state out at a time		
Smallest elasticity	0.337	0.366
Largest elasticity	0.641	0.792

Notes: This table reports the baseline specification of Table 1, columns (1) and (4), with three measures of precision. CV1 is the usual cluster-robust variance estimator. CV3 is the cluster jackknife, computed by re-estimating the model 51 times, each time deleting one state. The bootstrap p-value is from a restricted wild cluster bootstrap with Rademacher weights and 9,999 replications. The effective number of clusters is computed from the partial leverage of each state for the coefficient of interest, following Carter et al. (2017) and MacKinnon et al. (2023). Panel C reports the smallest and the largest elasticity obtained when one state is deleted, dividing each coefficient by the mean of the outcome in the full sample.

regressor would not. In the CPS, a test at the five percent level rejects 10.8% of the time when standard errors are clustered by state, 10.0% when they are clustered by state and month, and 11.0% when they are not clustered at all. The finer levels do not over-reject relative to the state level. This says that the problem is not correlation within states that the clustering fails to capture, but the asymptotic approximation itself with this many clusters of this shape. It is the reason I treat the bootstrap rather than the cluster-robust estimator as my primary measure of precision.

Panel C addresses a different question. If the identifying variation is concentrated in a few states, the estimate might be one state's result. I re-estimate the model 51 times, deleting one state each time. The elasticity never changes sign, and deleting the state with the largest share of the identifying variation raises the estimate rather than lowering it. The concentration therefore works against finding an effect rather than producing one.

Table A.2: Robustness of the Baseline Specification

	CPS		SIPP	
	Elasticity	(Standard error)	Elasticity	(Standard error)
Baseline specification	0.499	(0.267)	0.538	(0.276)
Adding industry × month effects	0.467	(0.264)	0.551	(0.272)
Excluding 2008 to 2013	0.593	(0.284)	0.374	(0.217)
Excluding cells at the minimum benefit	0.433	(0.248)	0.261	(0.190)
Excluding bins 1 and 2	0.466	(0.262)	0.261	(0.214)

Notes: Each row re-estimates the baseline specification of Table 1, columns (1) and (4), on a different sample or with an additional set of fixed effects. The outcome is the transition from employment to temporary layoff. The estimates are elasticities. Cluster-robust standard errors, clustered at the state level, are in parentheses. Cells at the minimum benefit are the combinations of state, half-year, and earnings bin in which the simulated benefit equals the state's minimum.

A.6 Robustness

Table A.2 reports the baseline specification under a set of alternatives.

Regressing each characteristic of the workers in the SIPP on the benefit, with state-by-bin and month fixed effects, I find no relation with base-period earnings, which the bins are constructed from, with age or education, or with the share of workers in construction and manufacturing. The share of white workers falls slightly with the benefit ($t = -1.9$), and the baseline specification absorbs race with its demographic fixed effects. The second row of Table A.2 adds industry-by-month fixed effects, which absorb the industry composition directly, and the estimate does not move.

The third row excludes 2008 to 2013, when Extended Benefits and the Emergency Unemployment Compensation program lengthened benefit durations in many states, so that the estimate does not rely on a period when the benefit system was changing in ways other than the weekly amount. The fourth row excludes the cells in which the simulated benefit equals the state's minimum, and the fifth excludes bins 1 and 2. In most states the minimum is the benefit paid at the lowest earnings that qualify, so that workers in these cells mostly earn too little to qualify. The estimates fall to 0.43 and 0.47 in the CPS and to 0.26 and 0.26 in the SIPP, between 1.2 and 1.8 standard errors above zero. Since these bins also have the highest replacement rates, where the model predicts the strongest response, the lower estimates need not reflect a bias.

Future changes in the benefit. If states changed their schedules in response to the layoffs of particular earnings groups, or if firms anticipated the changes, future benefits would predict the transition into temporary layoff today. I add to the baseline specification the change in the log benefit of the worker's bin over the following one, two, or three years. As elasticities, its coefficients are

0.34, 0.10, and 0.24 in the CPS and −0.36, −0.31, and −0.07 in the SIPP, with standard errors clustered by state between 0.26 and 0.49. The elasticity with respect to the current benefit stays between 0.46 and 0.58. In a distributed-lag event study with four half-year leads and four half-year lags (Schmidheiny and Siegloch 2023), the leads are jointly insignificant, with p-values of 0.42 in the CPS and 0.81 in the SIPP.

B Unemployment Insurance in the U.S.

UI in the U.S. is a joint state-federal program: each state runs its own program, within guidelines set by federal law (U.S. Department of Labor 2023).

Benefits. The weekly benefit amount (WBA) rises with the worker's earnings in the base period, between a minimum and a maximum that each state sets, and the replacement rate is the WBA divided by weekly earnings before the job loss. In many states the maximum is a percentage of the state's average weekly wage, and in a few the minimum is as well; elsewhere both are set in law. These rules differ across states, and each state changes them at its own times. Benefits last at most 26 weeks in most states.

Eligibility. A worker must have lost the job through no fault of the worker's own and must have earned enough in the base period, which in most states is the first four of the last five completed calendar quarters before the claim; most states require earnings in at least two of these quarters. Claimants must look for work, but most states waive this requirement for a worker on temporary layoff who can reasonably expect to be recalled soon. A worker who does not respond to a recall loses eligibility.

Financing. Benefits are financed by payroll taxes on employers, and in Alaska, New Jersey, and Pennsylvania also by contributions from workers. An employer's tax rate rises with the benefits collected by the workers it laid off, between a minimum and a maximum rate, so that experience rating is partial: in most states an employer at the minimum rate pays a positive tax even without layoffs, and one at the maximum pays nothing more for additional layoffs (Appendix D).

C The Calibration of Section 5: Details

Table 5 in Section 5.1 lists the parameters.

Value of non-work time. In the survey of Davis and Krolikowski (2025), 54.5% of the workers on temporary layoff would have accepted a temporary pay cut of 5% to avoid the layoff, and between 34% and 43% would have accepted cuts of 10% to 25%. A worker who would accept a cut prefers working at the wage to the layoff, which places the match among negative-gain matches. The share who would accept a 5% cut is therefore a lower bound on the share of negative-gain matches, since some of the others may only have found the cut too large. The twelve bins with the lowest replacement rates account for 52% of the transitions into temporary layoff, the step of one bin closest to 54.5%, and $\ell = 0.478$, the midpoint of the values that make exactly these bins negative-gain, matches this share. The calibration matches this lower bound rather than the share who would accept the cut itself: in the model, a worker accepts a temporary cut of 5% only if $b_k + \ell < 0.95$, which holds in the matches that account for 28% of temporary layoffs. Of the other shares in Table 7, the one near 32% is the midpoint between 41% and 24%, the shares of the workers on temporary layoff who would have accepted a cut of 5% and of 25% and do not say that it would have been too small to save the job.

Targets. The rates at which shocks arrive match the monthly probabilities of moving from employment to temporary layoff by bin. I take the ratios of these probabilities across bins from the sample of the main table and scale them so that the probability of moving from employment to temporary layoff across all employed workers is 0.41% per month, its value in the CPS; the sample of the main table, which requires earnings in the fourth interview, leaves out workers who were not employed at that interview and has a lower probability among workers who stay in the labor force, 0.28%. The benefit elasticity of the model is the response to a change of 1% in all benefits, measured between a fall and a rise of 1%. Because the response of a bin changes slope where its matches switch between the two types, and the bins in the middle of the earnings distribution lie close to that border, the share of joint decisions depends somewhat on the size of the change: it is 0.11 for an infinitesimal change and 0.12 for a change of 5%, close to the standard deviation of the log benefit after the fixed effects in bins 1 to 12.

Monthly transitions. A producing worker in a match of type k enters temporary layoff at rate H, which is $\lambda_d P_k$ times the share of producing workers who are in normal times, $1/(1 + n_k)$, where $n_k = \lambda_d(1 - P_k)/(\delta + \lambda_r)$ is the mass of matches producing through a shock relative to those in normal times: a shock that does not lead to a layoff lasts until it ends or the match is destroyed. The calibration matches the probability that a worker producing at one interview is on layoff at the next, which I approximate by $H\,[1 - \exp(-x_T)]/x_T$, where $x_T = \delta + \lambda_r + s$ is the rate

of leaving layoff.[14] With the rate of moving to a new employer fixed, the duration of layoffs does not depend on the benefit, and the share of producing workers who are in normal times makes the elasticity of this probability with respect to the benefit larger than that of P_k by a factor of $1 + n_k P_k/[(1 + n_k)(1 - P_k)]$, which lies between 1.003 and 1.025 across bins in the calibrated model. The same factor applies to the response to experience rating, so that it cancels from the ratio of the two responses in equation (26) within each bin.

Output. Output y sets the firm's value of a new match, averaged over earnings bins, to 1.30 months of output, or 1.35 months of the wage. Under free entry, this value equals the expected cost of a hire. In Blanco et al. (2026), the flow cost of a vacancy is K times the flow output of the match that it forms, a vacancy is filled at the rate $\theta^{-\alpha}$, where θ is market tightness, and an unemployed worker finds a job at the rate $f = \theta^{1-\alpha}$. The expected cost of a hire is therefore $K\theta^{\alpha}$ months of output. Their elasticity of the matching function is $\alpha = 1/2$, so that $\theta^{\alpha} = f$ and the expected cost is Kf months of output. With their $K = 2.962$, the monthly job-finding rate that their calibration targets, 0.44, gives 1.30 months of output, and the rate in their calibrated model, 0.50, gives 1.48 months.

With exponential shocks and both thresholds positive, a higher value raises the two thresholds by the same amount. The rates at which shocks arrive, refitted to the transition into temporary layoff in each bin, absorb the change. The share of joint decisions at the margin and the density of shocks there then do not depend on the value. The two responses move only through the cutoff and the shift of experience rating, which depend on the firm's values. Holding ϕ_+ and μ fixed, raising the value from 1.30 to 1.51 months of output, the largest value that the targets allow, moves the benefit elasticity and the response to experience rating by less than 0.001.

The pair's threshold is lowest in the lowest earnings bin, where the worker's gain from a layoff is largest. A low enough value pushes it to zero, where the joint decisions in that bin no longer respond: a calibration that sets the value to 0.64 months of output brings the pair's threshold in the lowest bin to zero and gives a share of joint decisions of 0.14.

Expected benefits. With benefits scaled by 0.46, as in Section 5.1, the density of shocks at the margin that matches the response to experience rating is about twice as high, and the targets cannot be matched at the baseline output. The recalibration therefore lowers output to 1.040, the largest value at which the rates at which shocks arrive can match the transitions into temporary layoff both

[14] The approximation holds the inflow constant over the month. The exact probability, which follows the worker through normal times, shocks, layoff, and joblessness over the month, is 1.0% lower in the calibrated model, a difference that the rates at which shocks arrive would absorb. At the calibrated parameters, it gives a benefit elasticity of 0.495 and a response to experience rating of −2.300, against the targets of 0.50 and −2.31; their ratio, which sets the share of joint decisions through equation (26), is 0.6% lower, which would move the share by less than 0.001.

in the calibrated economy and when the firm decides alone. This gives a value of a new match of 0.73 months of output, and no output brings it to 1.30 months; the share of joint decisions stays at 0.12. A firm deciding alone would then need the rate of moving to a new employer to fall by 0.16% for each 1% increase in the benefit for a zero response, and by 0.45% to reach the measured elasticity.

D The Experience-Rating Share

D.1 Construction

Tax schedules. States report their tax schedules each year to the Department of Labor in the ETA 204 Experience Rating Report. Section C of the report lists, for each interval of the experience factor, the average tax rate assigned to the employers in the interval, the number of employers, their total and taxable payroll, and the benefits charged to them. The experience factor is the benefit ratio in benefit-ratio states, the benefits charged to an employer over the past three years divided by its taxable payroll, and the reserve ratio in reserve-ratio states, the employer's balance of contributions less charged benefits divided by its taxable payroll. I use the rate years 1998–2019, in which 48 states report a usable schedule: 19 under a benefit-ratio system and 31 under a reserve-ratio system, with New Mexico and South Carolina under each system in different years. I make the average rate monotone in the experience factor with an isotonic regression, so that the rate rises with the benefit ratio and falls with the reserve ratio, and I measure the slope of the schedule between its minimum and maximum rates by weighted least squares.

Tax cost per dollar of charged benefits. Following Topel (1983) and Card and Levine (1994), the tax cost per dollar of charged benefits is the present value of the additional taxes that an employer pays for an additional dollar of charged benefits. In a benefit-ratio state with a three-year window, a dollar of charged benefits raises the benefit ratio by $1/(3W)$ in each of the next three years, where W is annual taxable payroll. This cost is therefore the slope of the schedule times $(1/3)\sum_{j=1}^{3}(1+i)^{-j}$. In a reserve-ratio state, a dollar of charged benefits lowers the reserve by a dollar, and the higher rate rebuilds it at the rate ζ, the absolute slope of the schedule. This cost is therefore $\zeta/(i+\zeta)$. Employers on the minimum or maximum rate face no such cost. I set the interest rate to $i = 0.05$ and taxable payroll growth to zero. The experience-rating share, which Card and Levine (1994) and Ratner (2013) call the marginal tax cost, is this cost times the charged share σ, the benefits charged to employers divided by the benefits paid to workers of taxable employers.

Table D.1 reports the share by system. Weighted by charged benefits, which weights the employers of laid-off workers, the share averages 0.47; weighted by taxable payroll, which weights

Table D.1: The Experience-Rating Share by Rating System

	Benefit ratio	Reserve ratio	All states
Charged share, σ	0.86	0.87	0.87
Tax cost per dollar of charged benefits, sloped part	0.87	0.75	0.79
Share, weighted by charged benefits	0.49	0.46	0.47
weighted also by benefits paid in the state	0.47	0.43	0.45
Share, weighted by taxable payroll	0.55	0.55	0.55
Charged benefits at the maximum rate	0.29	0.25	0.27
Charged benefits at the minimum rate	0.02	0.04	0.03
Taxable payroll at the maximum rate	0.05	0.05	0.05
Taxable payroll at the minimum rate	0.22	0.12	0.16
State-years	354	582	936

Notes: Means over the state-years 1998–2019 of the ETA 204 Experience Rating Report, each state-year weighted equally. The share is the tax cost per dollar of charged benefits times the charged share, weighted within a state-year by the benefits charged to each interval of the schedule or by its taxable payroll; the second row of the share also weights state-years by the benefits paid in the state. The tax cost on the sloped part is weighted by charged benefits. The last four rows are shares of charged benefits or of taxable payroll in the intervals at the maximum or the minimum rate.

the average employer, it averages 0.55. The difference comes from the maximum rate, at which 27% of charged benefits but only 4.5% of taxable payroll sit. An interest rate of 0.03 or 0.10 gives a benefit-weighted share of 0.51 or 0.41. For comparison, Ratner (2013) reports a share of 0.54 for 2001–2010 at an interest rate of 0.10, with 17.7% of employers at the minimum rate and 6.7% at the maximum in 2010, weighted by employment, and Card and Levine (1994) report 0.68 for 1979–1987.

Representative employers by sector. To obtain variation within a state and year, I follow Card and Levine (1994) and place the representative employer of each NAICS sector on its state's schedule at the sector's steady-state cost rate,

$$c_{sj} = \sigma_s \, u_{sj} \, B_s \, \frac{52}{W_{sj}},$$

the benefits charged per dollar of taxable payroll. Here u_{sj} is the insured unemployment rate of sector j in state s: the sector's share of UI claimants in the ETA 203 report, available from 2001, times the state's insured unemployment from the ETA 5159 report, divided by the sector's private employment in the Quarterly Census of Employment and Wages (QCEW). B_s is the state's average weekly benefit from the ETA 5159 report, and W_{sj} is the sector's taxable payroll per worker in the QCEW. All inputs are averaged over the three calendar years before the rate year, or over the years available at the start of the data, since the industry breakdown of the ETA 203 report starts

Table D.2: Representative Employers by Sector

NAICS	Sector	Insured unemployment (%)	Cost rate (%)	On a plateau	Share
11	Agriculture	5.86	6.87	0.41	0.41
21	Mining	2.99	3.13	0.08	0.62
22	Utilities	2.15	2.71	0.22	0.58
23	Construction	5.88	5.75	0.23	0.56
31-33	Manufacturing	3.19	3.79	0.08	0.68
42	Wholesale trade	2.13	2.41	0.10	0.66
44-45	Retail trade	1.53	2.05	0.06	0.70
48-49	Transportation	2.46	2.84	0.07	0.69
51	Information	2.66	3.29	0.16	0.62
52	Finance and insurance	1.54	1.83	0.03	0.73
53	Real estate	1.98	2.35	0.07	0.69
54	Professional services	2.28	2.63	0.14	0.65
55	Management	0.65	0.59	0.35	0.48
56	Administrative and waste	3.85	4.32	0.16	0.62
61	Educational services	2.88	10.15	0.55	0.32
62	Health care	1.22	2.55	0.03	0.72
71	Arts and recreation	2.56	3.93	0.11	0.67
72	Accommodation and food	1.39	1.92	0.07	0.70
81	Other services	2.17	3.18	0.07	0.69
All		2.36	3.09	0.11	0.66

Notes: Means over the state × rate-year cells 2002–2019, weighted by private employment in the QCEW. The insured unemployment rate and the cost rate are in percent of employment and of taxable payroll. "On a plateau" is the share of employment in cells whose representative employer is at the minimum or the maximum rate.

in 2001. In a benefit-ratio state the benefit ratio of the representative employer is its cost rate, and its tax cost per dollar of charged benefits is that of the interval of the schedule that contains it. In a reserve-ratio state the representative employer is on the sloped part of the schedule in the steady state if its cost rate lies between the minimum and maximum rates, and its tax cost per dollar of charged benefits is that of the slope.

Table D.2 reports the result for the 47 states and the rate years 2002–2019 in which every input is available. The share averages 0.66, close to the 0.68 of Card and Levine (1994). Sectors with high cost rates, such as agriculture and educational services, sit more often on the maximum rate and have lower shares, and management, with the lowest cost rate, sits more often on the minimum rate. Within a state and year the share has a standard deviation of 0.20 across sectors.

The share from schedule changes. The share of a sector moves with its cost rate, which reflects its own layoffs, and with its state's schedule. To use only the second source, I hold the sector's cost

rate and the state's charged share at their means over the rate years and let the share move only with the schedule. This share has a correlation of 0.70 with the share that lets the position move.

D.2 Robustness

The first stage on the tax side. If the measured share were the effective one, a sector on the sloped part of its schedule would see its tax rate rise one for one with its cost rate: the schedule slope in benefit-ratio states, about one, and one in the long run in reserve-ratio states. Table D.3 shows that the slope is about 0.25 in both systems. Measurement error in the cost rate can explain part of the shortfall. When the cost rate is instrumented with the layoff intensity of the sector in the CPS, whose errors come from a different survey, the slope rises to 0.46, while the cost rate three rate years earlier, which removes only transitory noise, leaves it at 0.28. Even after the correction, taxes respond to layoffs at less than half of what the schedules imply. The dispersion of employers within a sector explains part of the remainder: the share of the actual distribution of employers on the schedules is on average 0.77 of the share of the representative employers of the same states. This is why the main design measures the response per unit of the share that reproduces the response of the tax rate, rather than per unit of the share that the schedules imply.

The cost rate itself. In the baseline specification of Table 2, the coefficient on the cost rate itself is 0.02, with a standard error of 0.08: the tax rate responds to the cost rate almost entirely through the measured share, as the scale κ assumes. Without state-by-sector effects and in benefit-ratio states, it is 0.22 to 0.32, and part of the response does not run through the measured share.

Regressions on individual workers. Table D.4 adds the measured share to the regression of Section 4.3, with the log statutory benefit, the baseline fixed effects, and sector $\times$ year effects. The coefficients are per unit of the measured share; dividing them by the scale κ of Table 2, about 0.2, gives the response per unit of the effective share. The transition into temporary layoff falls with the share, by 0.12 log points with the full variation and by 0.25 with the share from schedule changes and state $\times$ sector effects, less than the 0.30 and 0.43 of the cells but within about one standard error of them. Adding the share leaves the benefit elasticity of the transition into temporary layoff at about 0.75 on this sample. I do not use the SIPP for experience rating: it covers only the rate years 2002–2013, in which the schedules change too little to separate their changes from the sectors' own layoffs.

Table D.3: The Response of Tax Rates to the Cost Rate

	Benefit ratio	Reserve ratio	All states
OLS, all cells on the sloped part	0.263	0.251	0.254
	(0.043)	(0.041)	(0.035)
Cells	5,120	8,399	13,519
OLS, cells with the CPS measure	0.273	0.250	0.259
	(0.043)	(0.041)	(0.035)
Cells	4,356	7,285	11,641
IV, CPS layoff intensity	0.440	0.482	0.465
	(0.057)	(0.064)	(0.047)
First-stage F	335	136	504
Cells	4,356	7,285	11,641
OLS, cells with the lagged cost rate	0.278	0.245	0.258
	(0.047)	(0.042)	(0.037)
Cells	4,249	6,872	11,121
IV, cost rate three rate years earlier	0.307	0.252	0.275
	(0.058)	(0.048)	(0.042)
First-stage F	3871	3934	7484
Cells	4,249	6,872	11,121

Notes: Regressions of the sector's assigned tax rate in the QCEW on its cost rate, both in percent, on the state × sector × rate-year cells, 2002–2019, whose representative employer is on the sloped part of the schedule. All regressions include state × year and sector × year effects and are weighted by private employment. The CPS layoff intensity is the monthly transition from employment to temporary layoff or permanent separation of CPS workers in the state and sector, averaged over the three calendar years before the rate year, in cells with at least 30 observations. Cluster-robust standard errors, clustered at the state level, are in parentheses.

Table D.4: Experience Rating in Regressions on Individual Workers in the CPS

	(1)	(2)
Log statutory benefit	0.76	0.75
	(0.46)	(0.46)
Measured share	−0.12	−0.25
	(0.12)	(0.13)
Share from schedule changes only		Yes
State × sector effects		Yes
Observations	1,777,625	1,777,625

Notes: Linear probability models of the monthly transition from employment to temporary layoff on the log statutory benefit and the measured experience-rating share of the worker's state, NAICS sector and rate year, with the fixed effects of the baseline specification of Table 1 (state × bin, bin × month, and demographics) and sector × year effects. Coefficients are divided by the mean of the outcome. The regressions use the CPS over 2002–2019. Survey weights are used. Cluster-robust standard errors, clustered at the state level, are in parentheses.